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A review of in-situ thermal monitoring techniques in polymeric powder bed fusion

Han Yu Ooi [a b], Jordan Noronha [a b], Tristan Pon [a b], Martin Leary [a b], David Downing [a b], Eric MacDonald [c], Stuart Bateman [a b], Raj Das [a b], Mahyar Khorasani [a b] Show more

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Outline

Abstract

In-situ thermal monitoring is a versatile tool in additive manufacturing, enabling real-time process control, recording for process certification and qualification, and the investigation of underlying process mechanisms. However, this field lacks a systematic review, especially in terms of applied technology. This review discusses thermal monitoring techniques in polymeric powder bed fusion, focusing on thermal imagers. Underlying principles of thermal imaging, their limitations and key considerations are presented. Existing academic usage in polymeric powder bed fusion is discussed, and tabulated information of thermal cameras and their specifications are presented. It was found Indium Antimonide medium-wave infrared (3–5 µm) sensors dominate the higher frame rate and resolution models, offering excellent thermal sensitivity (NETD), while cheaper microbolometer long-wavelength infrared (7–14 µm) sensors were more accessible but lower performance. Additionally, temperature-dependent emissivity was found to be highly significant in accurate data collection, however newer, more productive forms of polymeric powder bed fusion such as High-Speed Sintering, Large Area Projection Sintering and Multi Jet Fusion are largely unexplored thermographically. Further investigation of these newer processes thermo-optical properties has promise in better leveraging their high productivity through improved understanding of their underlying behaviour.

Keywords

Polymer powder bed fusion; Selective Laser Sintering; Multi Jet Fusion; Process monitoring; Thermal imaging

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Nomenclature

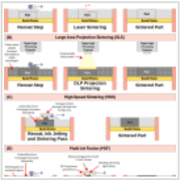
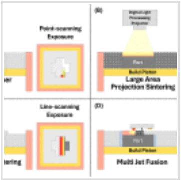
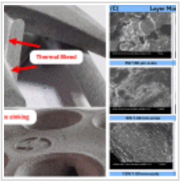
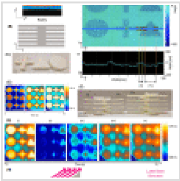
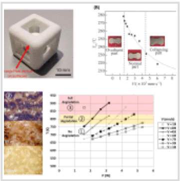
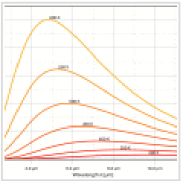
α-Si
: Amorphous Silicon

AM
: Additive Manufacturing

CCD
: Charge Couple Device

CB
: Carbon Black

Figures (18)

GB
Glass Bead

HSS
High Speed Sintering

HgCdTe
Mercury Cadmium Telluride

InSb
Indium Antimonide

LAPS
Large Area Projection Sintering

LWIR
Long-Wavelength Infrared

MJF
Multi Jet Fusion

ML
Machine Learning

MWIR
Medium-Wavelength Infrared

NETD
Noise Equivalent Temperature Delta

NIR
Near Infrared

OEM

Show 12 more figures

Tables (3)

Original Equipment Manufacturer

PA12
: Polyamide 12 / Nylon 12

PAEK
: Polyaryletherketone, family of polymers

PBF
: Powder Bed Fusion

PE
: Polyethylene

PEEK
: Polyetheretherketone

PEKK
: Polyetherketoneketone

PP
: Polypropylene

RTD
: Resistance Temperature Detector

SLS
: Selective Laser Sintering

SWIR
: Short-Wavelength Infrared

TPU
: Thermoplastic Polyurethane

VOx

Vanadium Oxide

1. Introduction

Additive manufacturing (AM) as a field offers immense promise towards more efficient usage of resources. Parts produced by AM are near-net shape, improving the buy-to-fly ratio of materials, with some parts amenable to usage directly after printing. AM facilitates greater shape complexity versus traditional manufacturing [1], [2], enabling more efficient structures, simplifying assemblies [3], [4], [5], and reducing material consumption. Similarly, waste byproducts and tooling requirements have the potential to be greatly reduced versus traditional manufacturing methods [3], [4]. Numerous families of AM exist, namely Binder Jetting, Directed Energy Deposition, Material Extrusion (commonly referred to as Fused Deposition Modelling or FDM), Material Jetting, Sheet Lamination, Vat Polymerization and Powder Bed Fusion (PBF) [6]. PBF is favoured by industry due to high geometric resolution and a wide range of processable materials [7].

AM of polymeric materials for industrial applications is dominated by the oldest form of PBF, Selective Laser Sintering (SLS), as well as FDM printers due to their low cost of entry and relative ease of use. Polymeric materials offer numerous benefits, namely lower density versus metals and ceramics, ease of processing partially due to lower melting temperatures, and low cost. Polymers cover a wide range of price-performance tiers: low-cost commodity plastic Polypropylene (PP) though uncommon can be printed and offers good chemical resistance, while higher performance engineering polymer Polyamides are the workhorse of polymeric PBF feedstock due to good mechanical properties and forgiving process windows [8], [9]. High temperature polymers such as the Polyaryletherketone (PAEK) family offer some of the best performance and highest operating temperature windows [10]. This report, though focusing on polymeric PBF more broadly, discusses both SLS, as well as newer more productive forms of PBF such as Large Area Projection Sintering (LAPS), High Speed Sintering (HSS), and Multi Jet Fusion (MJF). These processes move away from the traditional laser sintering approach of SLS, utilising various forms of radiation absorption modification or masking to sinter polymers. Though promising for higher productivity, their novelty has an associated smaller body of literature compared to SLS. Accordingly, characterisation of the underlying processes, in particular how their differing method of sintering impacts the induced thermal fields remains an open area of investigation.

Investigating thermal fields specifically has numerous justifications. Thermal fields during the build and cooldown cycles have a strong influence on the resultant mechanical properties, as well as geometric and dimensional accuracy. Developing understanding of a given process to predict the likely outcome prior to manufacture and/or assembly for risk mitigation is highly pertinent to commercialisation of technology, and subsequent development of understanding helps push wider adoption. With increasing levels of computational power, complex simulation work has become increasingly accessible. Numerical verification to ensure simulations reflect author intent is one critical step, however experimental validation is also highly pertinent to evaluate the accuracy of the underlying model assumptions. In the verification and validation process, in-situ thermal monitoring is a useful tool for the latter. Application of thermal monitoring is not solely confined to research applications. Many commercial PBF machines from the original equipment manufacturer (OEM) utilise variations of closed-loop thermal control to ensure process stability and repeatability [11].

For the scientist and engineer seeking to understand the causes behind variation in structural properties, process monitoring is essential to better understand how build thermal fields correlate to the final parts [12]. The data collected also has applications beyond just build-to-build optimisation, with potential for long-term process certification [13], [14], or pre-emptive build result predictions enabled by the data collected [15]. Precise thermal control is essential to part dimensional accuracy and mechanical properties [16], [17]. For example, yield strength, fracture strength and modulus tend upwards with greater energy density in Polyamide 12 (PA12), in-part due to reduced porosity increasing the stressed area [18], before diminishing as polymer degradation occurs [19]. This is attributed to lower melt viscosity for increasing melt temperature aiding powder coalescence [19].

Despite the benefits of collecting this data, the associated cost is high. Uncooled, low-frame-rate and low-sensitivity Long-Wavelength Infrared (LWIR) bolometric sensors can cost below USD$1000, however research-orientated Medium-Wavelength Infrared (MWIR) sensors can range from USD$10,000 to USD$200,000 [20]. It is therefore important to understand the type of data the user wishes to collect, and the associated parameters of a thermal imager which thus becomes significant.

Existing reviews typically focus on in-situ monitoring more broadly, including optical techniques [21], [22], often being specific to or largely focused on metallic PBF [23], [24], or covering numerous process families (e.g. vat polymerization, sheet lamination, fused deposition modelling) [11], [25]. This paper is specific to polymeric PBF, noting the bulk of literature available focuses on SLS, but discussion of how requirements differ when investigating processes such as MJF, HSS, and LAPS is also presented. Minimisation of scope thus enables a more detailed discussion of considerations and concerns associated with or specific to polymeric PBF. Evaluation of existing academic usage of thermal imagers in polymeric PBF is presented, with the aim of making general recommendations for the class and family of thermal imager appropriate for typical applications in polymeric PBF monitoring.

2. Overview of polymeric powder bed fusion

The working definition of PBF is covered by ISO/ASTM 52900, which defines PBF more generally as an “additive manufacturing process in which thermal energy selectively fuses regions of a powder bed” [6]. PBF processes are characterised by the use of powder material feedstock. Typically, the build chamber has a powder reservoir from which powder is dispensed from into precisely controlled thickness layers. These layers are fused into geometric slices of the desired 3D body, building up in height from the base. Temperatures in the machine are typically controlled to minimise the temperature gradient in the part, inhomogeneous cooling resulting in internal stresses and subsequent dimensional inaccuracies in the final part [26]. The bulk of literature has historically focused on SLS, one of the earliest forms of AM. Newer, less researched, but equally promising processes such as LAPS, HSS, and MJF are also discussed. Key differences are shown in Fig. 1.

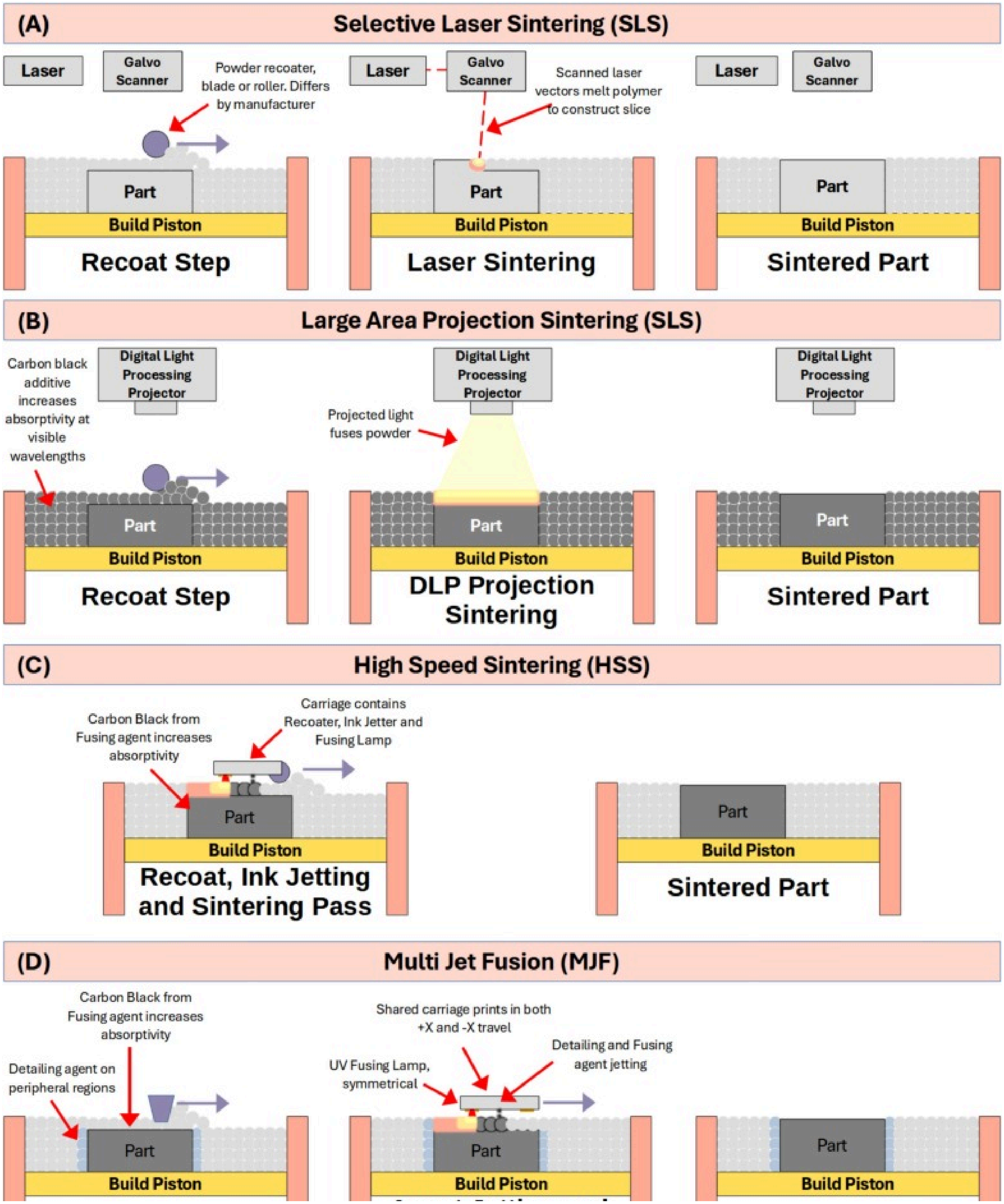

(A) Selective Laser Sintering (SLS)
Laser
Galvo Scanner
Powder recoater, blade or roller. Differs by manufacturer
Part
Build Piston
Recoat Step
Laser
Galvo Scanner
Scanned laser vectors melt polymer to construct slice
Part
Build Piston
Laser Sintering
Laser
Galvo Scanner
Part
Build Piston
Sintered Part
(B) Large Area Projection Sintering (SLS)
Carbon black additive increases absorptivity at visible wavelengths
Digital Light Processing Projector
Part
Build Piston
Recoat Step
Digital Light Processing Projector
Projected light fuses powder
Part
Build Piston
DLP Projection Sintering
Digital Light Processing Projector
Part
Build Piston
Sintered Part
(C) High Speed Sintering (HSS)
Carbon Black from Fusing agent increases absorptivity
Carriage contains Recoater, Ink Jetter and Fusing Lamp
Part
Build Piston
Recoat, Ink Jetting and Sintering Pass
Part
Build Piston
Sintered Part
(D) Multi Jet Fusion (MJF)
Carbon Black from Fusing agent increases absorptivity
Detailing agent on peripheral regions
Part
Build Piston
Shared carriage prints in both +X and -X travel
UV Fusing Lamp, symmetrical
Detailing and Fusing agent jetting
Part
Build Piston
Part
Build Piston

Fig. 1. Representative process diagrams of polymeric powder bed fusion technologies. (A) Selective Laser Sintering, (B) Large Area Projection Sintering, (C) High Speed Sintering, (D) Multi Jet Fusion.

2.1. Selective Laser Sintering

SLS is the earliest referenced literature on PBF of polymeric materials, first demonstrated by Deckard in 1986 [27], with the associated technology patented in 1987 [28]. Deckard's process describes the usage of a scanning laser to form sequential, discrete slices of a 3D model in a powder bed. SLS is perhaps the best understood form of polymeric PBF, with a plethora of research on thermal behaviour [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], imaging [12], [40], [41], [42] and subsequent material properties [12], [18], [33], [37], [43], [44], [45] to name a few.

SLS is capable of processing a broad range of materials. For example, flexible elastomeric Thermoplastic Polyurethane (TPU), commodity polymers Polyethylene (PE) and PP, and composite powders with a variety of additives [46]. In a recent review [9], engineering polymer PA12 were found to be the most studied feedstock (accounts for ~45% of literature), followed by high-temperature thermoplastic Polyetheretherketone (PEEK, ~12% of literature), and elastomeric thermoplastic polyurethane (TPU, ~5% of literature).

Collectively, high temperature and engineering polymers, defined as melting above 260°C and 140°C respectively, accounted for 76% of literature reviewed [9]. Accordingly, higher energy levels, and thus faster heating rates are required to maintain good bulk sintering rates for these polymers. Sintering significantly below melting has been demonstrated for high-temperature polymers such as Polyphenylene Sulphide (PPS) [47], however these approaches though beneficial do not fully address the fundamentally highly transient heating associated with the process. For example, Peyre et al. [17] demonstrated approximately 95% densification of PA12 powder, with a scan speed of 200 mm/s, power of 1.6 W, and spot size of 125 μm.

Rapid heating of the polymeric powder associated with high scan speeds drives highly transient melt kinetics [39]. Imaging of this behaviour accordingly requires high-frame rate devices to accurately capture. Due to the line-raster approach of SLS, the productivity is relatively slow as the laser spot size and associated track-width is small.

2.2. High throughput powder bed fusion of polymeric materials

High throughput forms of polymeric PBF are of interest due to the higher productivity for the end user. Of particular interest to this review are LAPS, HSS and MJF. Table 1 shows the key differences between each process, while Fig. 2 shows the difference in exposure methods between techniques.

Table 1. General comparison of polymeric Power Bed Fusion processes.

	SLS	**LAPS**	**HSS**	**MJF**
Source of Fusion	Galvanometer scanned laser (typ. CO_2)	DLP projected visible light	Infrared lamp, radiation modifying ink	Infrared lamp, radiation modifying ink
Transient Heating	Very high	Very low	Low	Low
Commercial OEM	Several, e.g. 3D Systems, EOS, Farsoon	Ascend Manufacturing	Stratasys, voxeljet	Hewlett-Packard

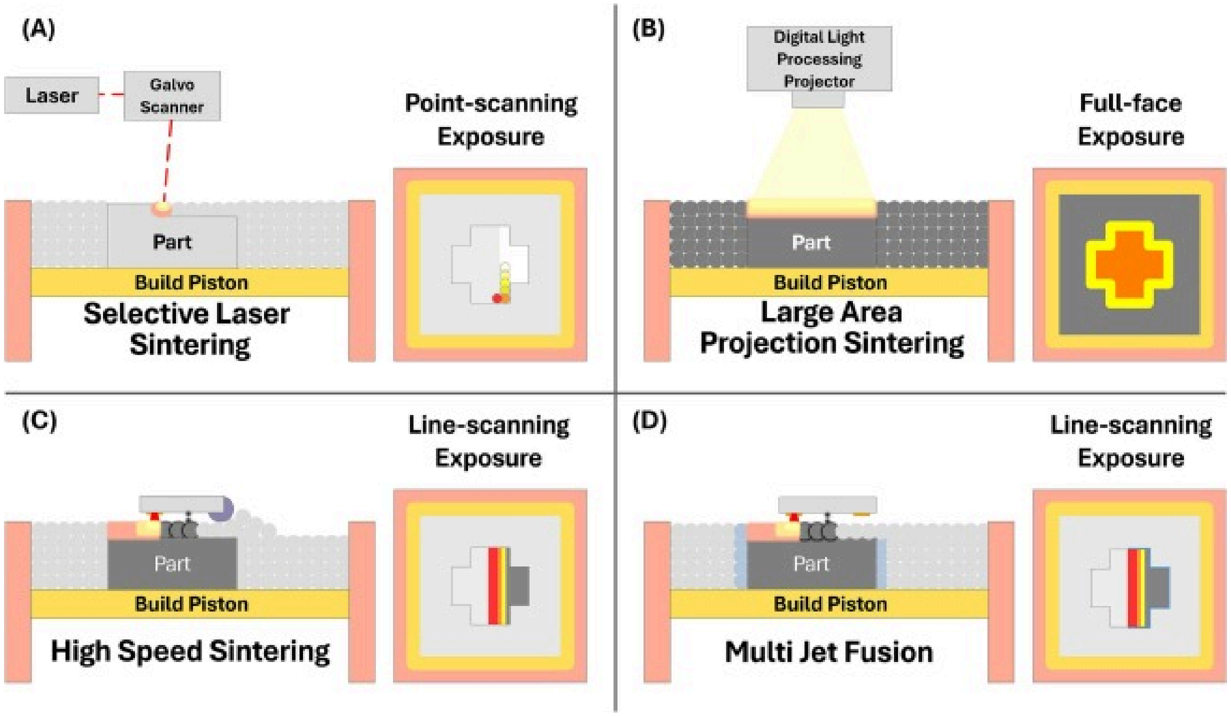


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Fig. 2. Exposure strategy of polymeric powder bed fusion techniques: (A) Selective Laser Sintering: point-scanning laser, (B) Large Area Projection Sintering: visible light projection with full-face exposure, (C) High Speed Sintering: line-scanned infrared exposure on radiation modifier masked regions, (D) Multi Jet Fusion: line-scanned infrared exposure on radiation and inhibiting agent masked powder bed.

2.2.1. Large area projection sintering

LAPS is one alternative to SLS, utilising a high-power projector to rapidly produce parts [7]. Invented by Gardiner [48], [49], the prototype used a high-resolution digital light processing (DLP) projector to print layer slices in one step. The more gradual heating of entire part cross-sections reduces the local heating rates versus some SLS machines by four orders of magnitude. In the context of in-situ thermal monitoring, this eliminates much of the frame rate requirement in thermal field imaging as there is no highly transient melt pool formation and kinetics present.

2.2.2. High speed sintering

HSS is another high productivity process which uses a radiation modifying agent to selectively increase the infrared absorptivity of the powder layers, the coated regions being subsequently fused with an infrared lamp [7], [50]. Developed by Hopkinson and Erasenthiran, and patented in 2003 [51], HSS is similar to LAPS in that the process has much lower heating rates. Thus, monitoring frame rate requirements are minimal. As the radiation agent modifies the powder absorptivity, additional complexity is introduced as thermal imagers are typically calibrated to a single fixed emissivity value.

2.2.3. Multi jet fusion

MJF is a proprietary process by Hewlett-Packard (HP), utilising ink-jetted detailing and fusing agents in conjunction with several infrared lamps to fuse polymer powder layers [7]. Similar in function to HSS, MJF selectively increases the absorptivity of the powder where fusing is desired through application of the absorptivity increasing fusing agent. Additionally, the detailing agent helps prevent sintering in regions while improving surface finish, typically deposited on the edges of part cross-sections. Much like LAPS and HSS, monitoring requirements are lower due to the absence of highly transient heating. Selective addition of the fusing and detailing agents which also result in modified emissivity of the powder adds increased complexity to non-contact thermal monitoring. Thermal monitoring typically relies on fixed emissivity calibration, academic investigation of this is largely unexplored as of this time.

2.3. Polymer crystallinity

Polymers are chain-like molecules formed from covalently bonded monomers [52]. Polymers may be differentiated as being amorphous (or non-crystalline), semi-crystalline, or crystalline. Crystallinity (denoted as a %) represents the degree of crystalline microstructure in a polymer, with the remainder being in an amorphous state [53]. Due to the increased order of crystalline morphologies, an associated volumetric shrinkage occurs during cooling [53]. This results in an increase in modulus as the more densely packed molecules make the crystal stiffer [53]. It has been shown experimentally [54] that temperature profile directly affected the degree of crystallinity in PA12 MJF.

As the extent and rate of crystallisation are key properties in determining dimensional inaccuracy in sintered parts, polymers with little or no overlap between the melting and recrystallisation temperature ranges suffer less geometric distortion from solidification shrinkage [7], [19]. Slow recrystallisation is also preferred for this reason [55], as simultaneous shrinkage across the part minimises distortion. Differential scanning calorimetry (DSC) characterisation of polymers struggles to directly quantify the exact rate of crystallisation, however the difference between melt onset temperature and recrystallisation onset temperature is directly related to the rate of crystallisation [55]. Accordingly, the larger the difference between the two temperatures, the lower the rate of recrystallisation and thus the lower the distortion observed in fabricated parts [55].

3. Thermal defects in polymeric powder bed fusion

Key drivers behind process monitoring more broadly are observation of defect generation in real time. Accordingly, defects specifically driven by thermal fields are introduced. A brief discussion of terminology and relevant information to frame subsequent discussion on thermal monitoring techniques is also presented. Defects listed occur in every form of polymeric PBF discussed, however the susceptibility of a process to a given defect will vary.

3.1. Thermal bleed

Thermal bleed or growths shown in Fig. 3 (A) occur when powder unintentionally adheres to the sintered part due to excess heat accumulation [55], [56]. This is more common for smaller particles due to the lower mass overheating more readily [55]. Geometric conditions of a given part impact the thermal field homogeneity, for instance overhanging surfaces printed on powder as opposed to previously solidified material can demonstrate higher peak temperatures due to worse thermal diffusion [57].

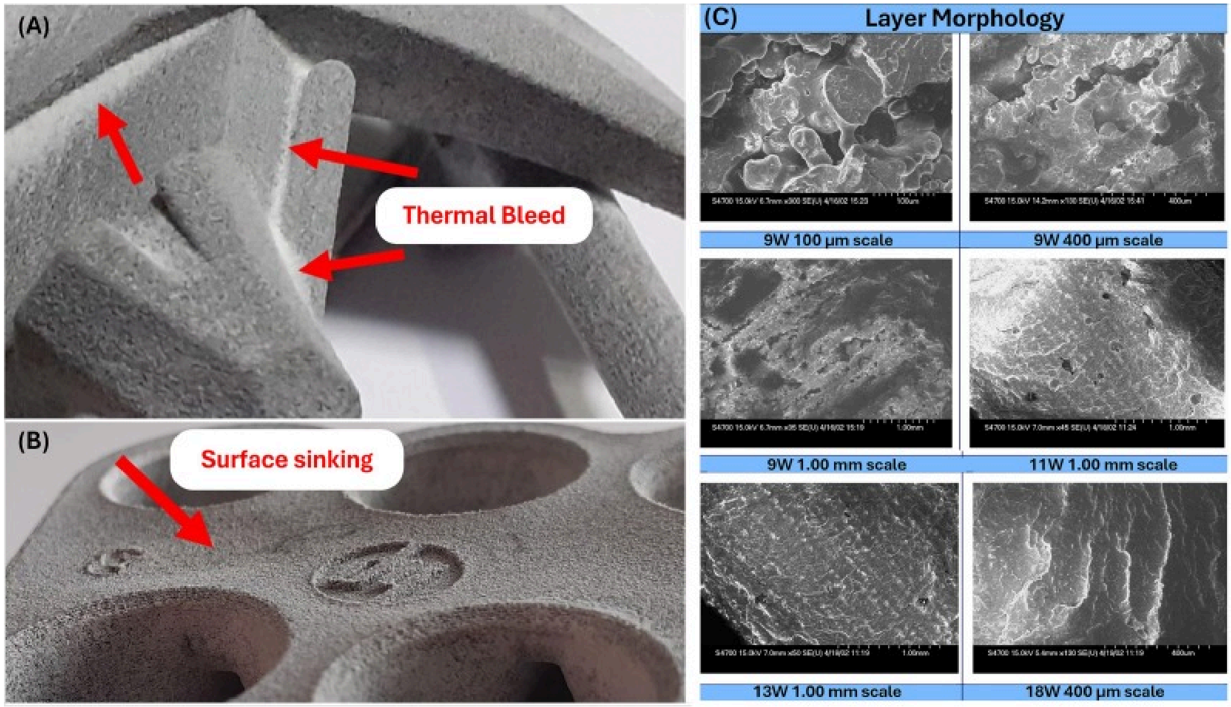

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Fig. 3. Representative thermal defects in polymeric PBF parts: (A) Unfused powder adhesion to part from thermal bleed, (B) Surface sinking of MJF PA12 from shrinkage [56] (Permission No: 6123390756793). (C) Layer morphology of Duraform Polyamide at different resolutions by varying laser power [18] (Permission No: 6123381371758).

3.2. Surface sinking

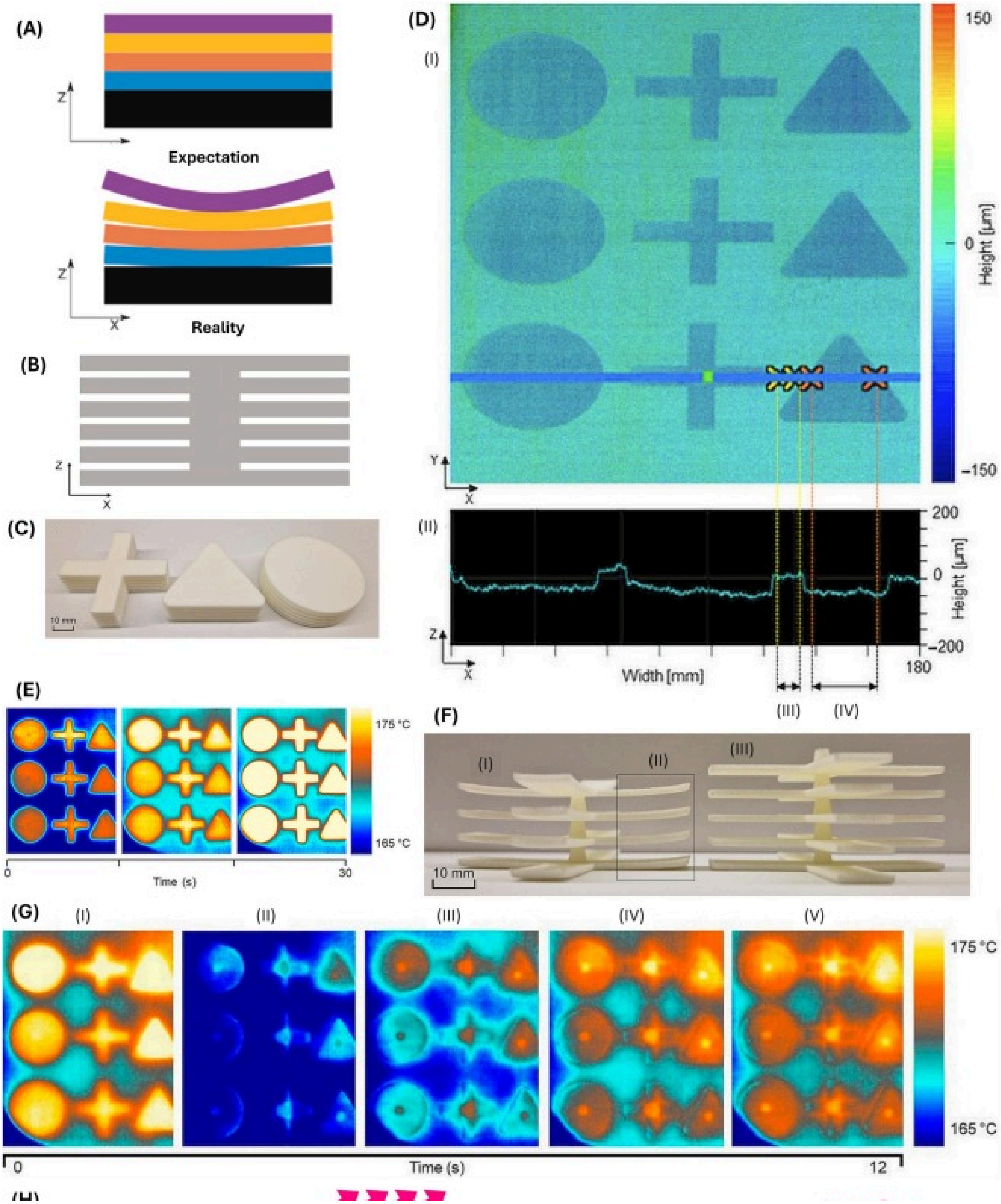
Surface sinking shown in Fig. 3 (B) occurs due to inhomogeneous part cooling, resulting in localised shrinkage of a surface's central region. This predominantly impacts upward-facing surfaces, as well as regions with high volume to surface ratios [56]. These attributes can be observed in thermography, and numerous control schemes may be applied to better optimise the input power and scan parameters to minimise their effects in SLS [58].

3.3. Porosity and lack of fusion

Porosity in SLS parts induce significant scatter in the material properties of the produced part [59]. Fig. 3 (C) shows the layer morphology at different powers, noting the 100 μm image at 9 W shows the voids between fused grains [18]. Anisotropic properties are typical, for example, elongation at break is particularly affected [60] as the density of voids (and thus stressed area) varies between the X, Y and Z axes. Incomplete particle melting (lack of fusion) is one driver of porosity, increasing as energy density decreases [59]. In addition, porosity is also driven by failure of particles to fully coalesce, as lower temperature melt possesses higher viscosity [61]. Porosity decreases mechanical properties by decreasing the stressed area of the part, presenting crack initiation sites.

3.4. Shrinkage, warpage and curling

Curling defects shown in Fig. 4 (A and F) occur due to residual stress from inhomogeneous cooling and thermal gradients in parts [14], [26]. As stresses are larger perpendicular to the scan direction and machine Z-axis (see Fig. 4 (H)), different exposure strategies can mitigate the deformation by minimising the maximum stress values and direction [62]. Heating of the powder substrate plate to reduce residual stress through reduction of the thermal gradient is a common strategy.


(A)
Expectation
Reality
(B)
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Height [µm]
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Width [mm]
180
(III)
(IV)
(E)
175 °C
165 °C
Time (s)
0
30
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(V)
175 °C
165 °C
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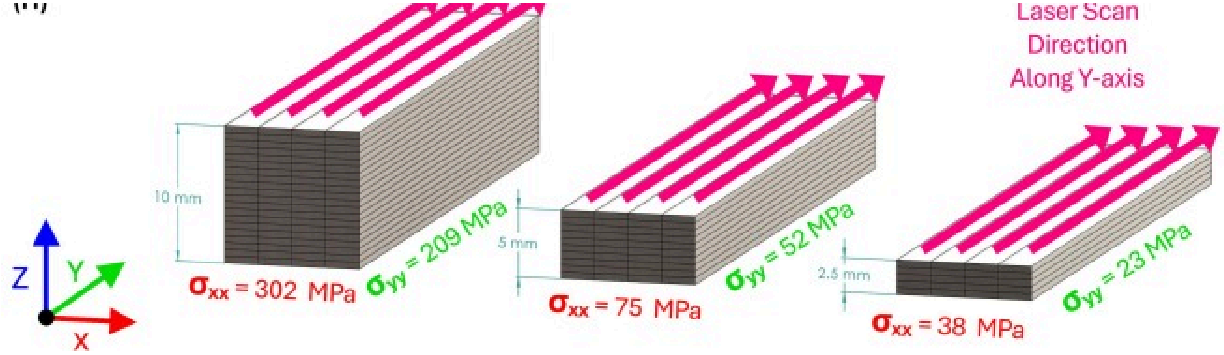

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Fig. 4. Curling in polymeric PBF: (A) Ideal layer configuration versus curled layer configuration [14] (open-access article CCBY4.0). (B/C) Part geometry of printed samples, (D) consolidation shrinkage from void reduction, (E) thermogram of undistorted layer during recoat, post-recoat and during sintering, (F) Printed part geometry: I/II) warped geometry with intentionally reduced process temperature, III) undistorted geometry, (G) progression of recoat induced curling: I) post-sintered layer showing good uniformity, ii) recoat thermogram, III/IV/V) curling apparent on thermogram [26] (open-access article CCBY4.0). (H) Residual stress in selective laser sintered/melted stainless steel 316L (whole of sample) using X-ray diffraction (Adapted from [62]).

Introduction of cold powder can also induce curling defects, with recoat compaction strategies offering some mitigation of this effect for in-situ curling [63]. Preheating of the powder in the recoater feeder is a typical strategy to minimise the thermal gradient between recoated powder and the last sintered layer, however excessive heating of the powder has been observed to increase powder cohesiveness [14]. Testing by Sillani et al. [14] showed improved part surface roughness for increasing recoat powder temperatures, however this trend was reversed for isotactic copolymerised PP, which also demonstrated increased powder cohesion clumping as recoat powder temperatures increased. Process and feedstock variability likely has a strong impact on the optimal temperature, highlighting the importance of characterizing novel and atypical setups.

Mousa [64] printing PA12 with varying percentages of glass bead filler (GB) found powder bed thickness was the most significant contributing factor to curling. Other significant factors in order of importance were part bed temperature, GB fill ratio, layer time, and laser power. More generally, polymer/filler systems suffer less from shrinkage. Part of this can be attributed to the percentage of polymer contributing only part of the overall volume, the total reduction from polymer consolidation will be proportionally less versus unfilled powder [16], though this assumes the filler does not exhibit significant shrinkage at the polymer sintering temperature. They also found that tensile modulus increases monotonically for increasing volume fractions, and tensile strength and failure strain increase with decreasing bead diameter.

Achieving good dimensional accuracy from SLS of amorphous polymers can in part be achieved through partially consolidating powder [16], though such parts typically possess notably poor mechanical properties. Semi-crystalline polymers such as the ubiquitous Polyamide family can be fully densified insofar as the melt viscosity is sufficiently low at temperatures below the degradation point [16], and the polymer remains at a sufficiently elevated temperature long enough to facilitate densification [17]. Parts are ideally maintained in a supercooled liquid state during the build process, where the part is incapable of maintaining internal stresses. Cooldown of the part at the end of the build can thus maintain more homogenous shrinkage, minimising warpage [16].

Aside from scan-strategy influencing the direction and magnitude of residual stress, it is evident most curling defects eventuate from thermal field inhomogeneities. As such, predicting these effects is key to minimising wastage and reducing iteration cost.

3.5. Powder degradation

Excessive temperature can cause depolymerisation and degradation of polymer [53]. For instance, used PA12 powder exhibits defects such as uneven part surfaces referred to as “Orange Peel” [30], [65] shown in Fig. 5 (A). Fig. 5 (B) shows the result of varying scan speed on a composite powder, the part transitioning from experiencing thermal degradation, to lack of fusion. Fig. 5 (C) is a more visible example of degradation in Polyetherketoneketone (PEKK) by Peyre et al. [17], excess power resulting in partial or full degradation with the powder showing black spots on the sintered surface.

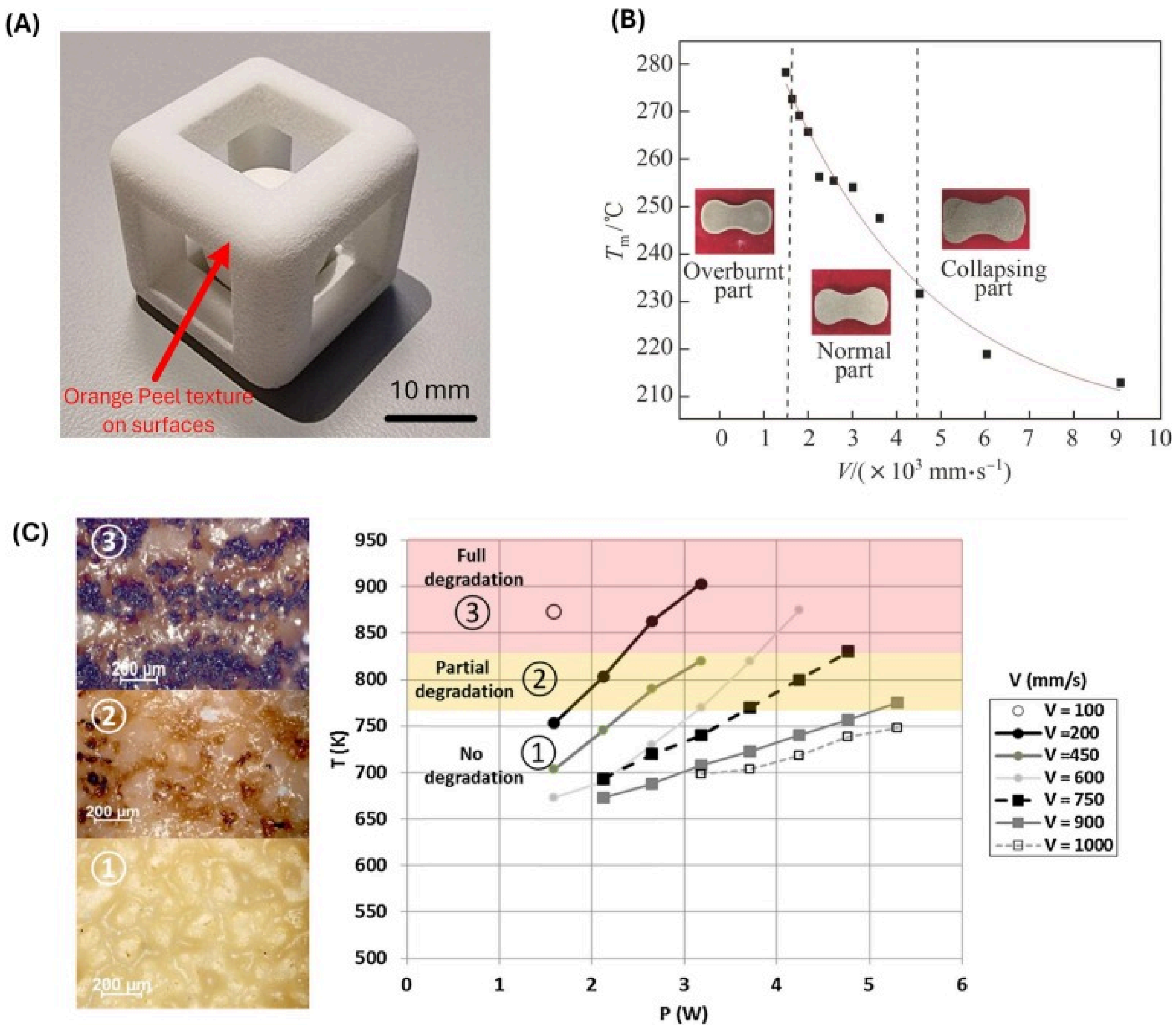

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Fig. 5. Overheating defects and characterisation: (A) Orange Peel on PA12 (PA2200) SLS part [68] (open-access article CCBY4.0). (B) Temperature and condition of coated sand part with varying scanning speeds, 200W Laser Power [69] (Permission No: 6123401402170). (C) Damage thresholds and temperature of PEKK at varying power and scan speeds [17] (Permission No: 6123410942242).

Chemical degradation associated with polymeric PBF includes non-exhaustively: thermal oxidation, solid-state post-condensation, molecular chain scission, molecular cross-linking and thermal property changes [66]. Changes in molecular weight also occur, detrimentally impacting powder flowability. Fully densifying amorphous polymeric parts typically requires parts to be heated well above the glass transition temperature (Tg), resulting in material degradation from local overheating [16]. Accordingly, amorphous polymeric parts can be intentionally partially consolidated, yielding good dimensional accuracy at the expense of aforementioned poor mechanical properties. Semi-crystalline materials benefit mechanically from the lower viscosity achieved through increased melt temperature aiding coalescence to a point, before degradation impacts occur [18], [19].

Polymer chain length growth, or post-condensation, can result from laser exposure and may be noted by higher viscosity versus the original powder for PA12 [67]. This is opposed to chain scission which will result in lower viscosity for a given temperature. Shifts in molecular weight distribution are negatively impacted by build temperature and time, while increased molecular weight in PA12 can be attributed to post-condensation reactions [30]. This also decreases the recrystallisation temperature.

These impacts affect both fused and unfused powder, limiting the degree of re-use, and necessitate reintroduction of varying percentages of virgin powder. Used PA12 powder in SLS suffers from worse surface morphology and higher roughness compared to virgin powder, while the melt distribution also increases [66]. Minimising degradation of the powder through precise thermal control limits powder wastage.

4. Thermal imaging

Thermal infrared imagers are broadly popular for non-contact in-situ monitoring, offering higher spatial and temporal resolution compared to pyrometers, though they have limitations in deriving absolute temperature [70]. Despite their cost, the quality of data collectable drives widespread usage. Though relatively prolific in usage, research in polymeric PBF rarely discusses the implementation details and uncertainty associated with thermal imaging.

Sensor selection and implementation are highly dependent upon what the user wishes to evaluate. Of the polymeric PBF processes, SLS has the most difficult requirements due to the laser heating powder rapidly, study of the highly transient melt-pool formation and kinetics requiring extremely high-frame rate sensors with sufficient resolution to prevent aliasing effects which reduce the apparent temperature of the measured geometry. LAPS, HSS and MJF do not exhibit highly transient behaviour due to the simultaneous, whole-of-layer sintering approach, thus low-frame-rate, high-resolution imaging is more appropriate. For many SLS users, collection of bulk temperature only can be sufficient however, eliminating high-frame-rate requirements and the high associated costs.

Before investigating the identified sensors in polymeric PBF literature, discussion of the underlying principles and two relevant sensor technologies applicable to polymeric PBF is necessary. Methodological error can be significant in thermal imaging, with a good understanding of the thermo-optical properties of the studied material required to measure absolute temperature. Regarding sensor categories, cooled photon detectors cover the high-performance, high-cost segment of the reviewed literature, offering excellent frame rate, sensitivity, and resolution. Uncooled microbolometer detectors are much cheaper; however, they suffer from limited maximum frame rate and often poorer thermal sensitivity. Numerous application notes should be considered before implementing or interpreting thermal imager data; however, the main considerations are discussed here.

4.1. Thermal infrared fundamentals

Typical temperature ranges for polymeric PBF processes range from room temperature (23°C or~300K), up to around 400°C (~670K) for high-temperature polymers such as Polyetherimide, Polyphenylene Sulphide, or the PAEK family [61]. PA12 is typically sintered around 180–200°C [71], while isotactic PP melts around 150–160°C [8]. Non-contact thermal monitoring techniques rely on Planck's law to measure the emitted radiation from an object to estimate temperature. Fig. 6 shows the emitted blackbody radiation by wavelength for the typical polymeric PBF temperatures, based on ISO9288:1989 [72] using the black body spectral exitance equation $M_\lambda^\circ\ (Wm^{-2}\mu m^{-1})$ given by Eq. (1), where variables λ is the wavelength and T is the temperature in Kelvin. Constants used are the Planck Constant h, Boltzmann constant k, and the speed of electromagnetic waves in a vacuum c_0. Here ° denotes blackbody calculations.

$$M_\lambda^\circ(\lambda,T)=\frac{2\pi hc_0^2}{\lambda^5\exp\left(\frac{hc_0}{k\lambda T}\right)-1} \tag{1}$$

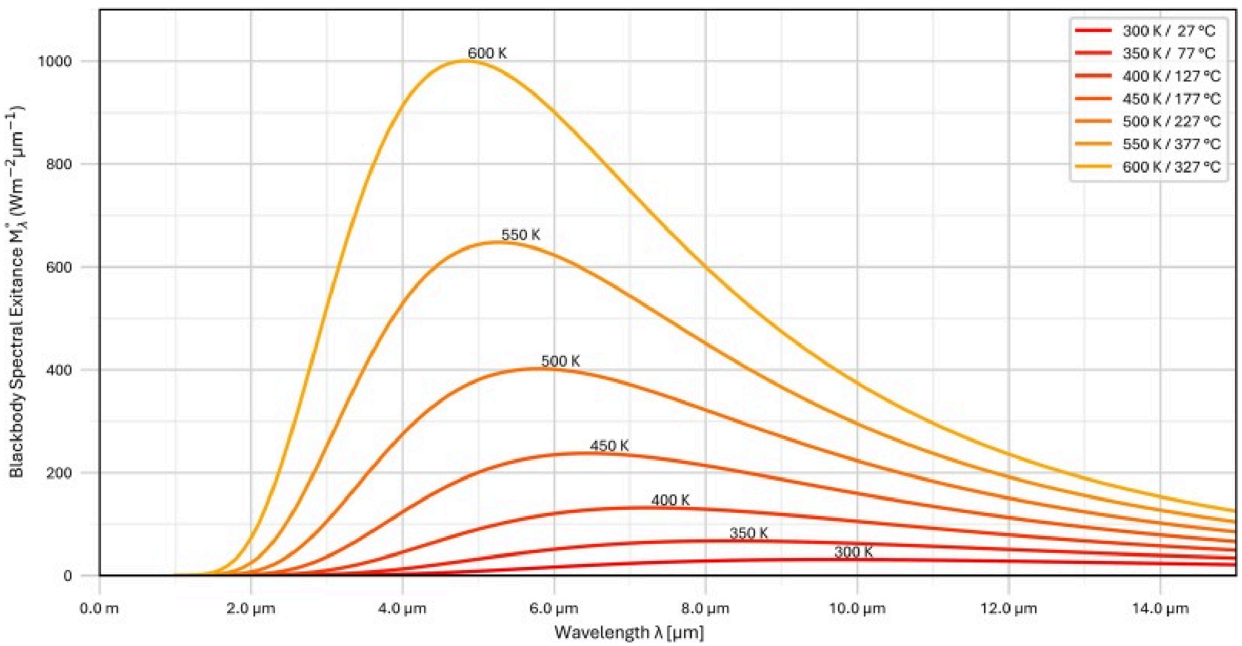

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Fig. 6. Emitted radiation for typical polymeric PBF temperatures by wavelength [72].

Calculating the wavelength at which emissions are the highest for a given temperature is calculated per Eq. (2), where T is the temperature in Kelvin.

$$\lambda_{\mathrm{m}} = \frac{2.898}{T}\ mm \tag{2}$$

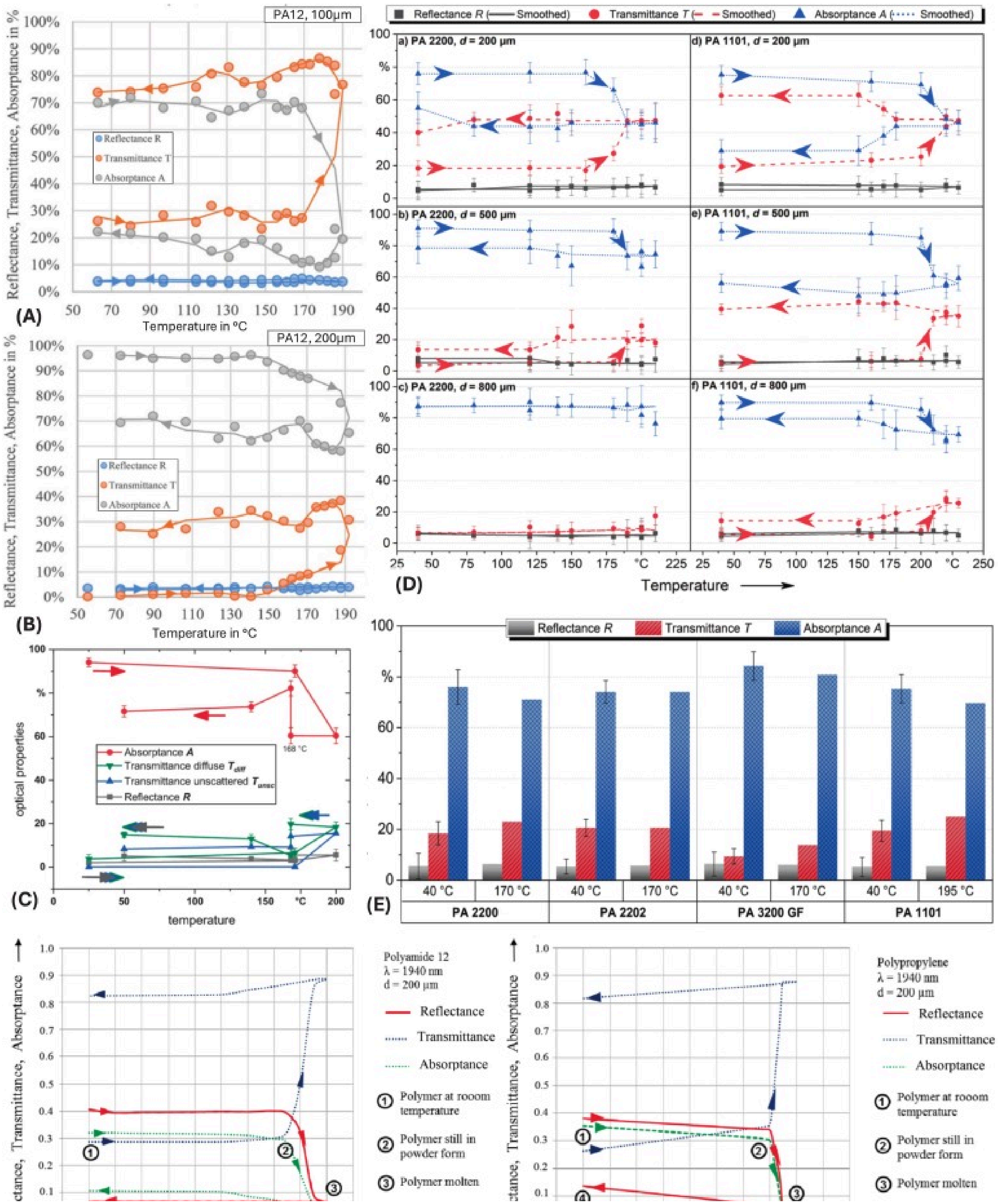
Real parts seldom behave like a perfect blackbody. Emissivity ϵ is a constant used to represent the proportion of radiation emitted with reference to a perfect black body, where ϵ ranges from 0 to 1. Graybody spectral exitance M_{gb} is approximately $\epsilon \times M_{\lambda}^{\circ}$ [40]. In practice, emissivity is also dependent upon the roughness, phase and temperature of the material, which results in systematic measurement inaccuracies reporting a single figure [39], [73]. Additives such as carbon black (CB) also increase emissivity [16] greatly across the spectral range, which may introduce additional complexity for HSS and MJF due to emissivity varying between sintered and unsintered regions.

It should be noted that although emissivity and absorptivity are equivalent same wavelength, academic usage varies frequently depending upon context. Generally, emissivity is used when discussing radiative heat rejection and emissions, while absorptivity is used in the context of energy input. Transparent materials may have higher apparent absorptivity (measured) than emissivity due to transmitted radiation being absorbed in sub-surface regions [74]. As such, this paper uses them interchangeably, typically based off the cited literatures usage, though this is not guaranteed. Another reason for emissivity and absorptivity to be given separately in literature is due to differing wavelengths. For example, a monitoring sensor may have one spectral sensitivity range (e.g. 3.0–5.0μm), while the sintering laser operates at a different wavelength (e.g. 10.6μm for a CO_2 laser).

4.2. Temperature and phase dependent emissivity

Emissivity shifts due to temperature and phase-change are demonstrably significant in experimental testing. This occurs due to changes in the microstructure of the material as it heats up and cools down. Yuan et al. [75] measured warmup, sintering and cooldown temperature fields with a 320×240-pixel, 60 Hz FLIR A325. The sample rate was 0.5 Hz for heat up and cooldown, and 2 Hz during sintering. Measuring in the 8–12 μm range, the sensor had a thermal sensitivity of 30 mK, and an absolute accuracy of 2 °C for known emissivity between the temperature range of 0–500 °C. Despite calibrating the sensor with a known blackbody, implausible temperatures around 300–400 °C were observed directly after the sintering scan, attributed to emissivity changes in the melted and partially melted polymer.

The depth of the sintered layer is significant for modelling energy absorption and thus absorptivity/emissivity ϵ. Typically, the exponential Beer-Lambert extinction law is applied to describe energy attenuation in the powder which is semi-transparent [76]. Experimental testing with double integrating spheres to quantify the shift in absorptivity of PA12 (EOS PA2200) and related powders was conducted by [71], [74], [77], [78]. Fig. 7 (A and B) show the effect of layer thickness on PA12 sintered with a CO_2 laser at 10.6 μm. Fig. 7 (C) shows good agreement with (A and B). Fig. 7 (D) shows the difference between EOS PA2200 (PA12) and PA1101 (PA11), while Fig. 7 (E) additionally shows the effect of CB additive (PA2202) and glass fibre additive (PA3200). Fig. 7 (F and G) shows the optical properties of PA12 and PP at a lower wavelength, 1.94 μm. Heinl [74], Schuffenhauer [71], and Marschall [77] found low reflectance at the 10.6 μm wavelength CO_2 lasers operate in, with almost no change irrespective of additive, temperature, or phase-change.



PA12, 100µm
Reflectance R
Transmittance T
Absorptance A
Reflectance, Transmittance, Absorptance in %
Temperature in °C
(A)
PA12, 200µm
(B)
Reflectance R (— Smoothed)
Transmittance T (– – Smoothed)
Absorptance A (····· Smoothed)
a) PA 2200, d = 200 µm
b) PA 2200, d = 500 µm
c) PA 2200, d = 800 µm
d) PA 1101, d = 200 µm
e) PA 1101, d = 500 µm
f) PA 1101, d = 800 µm
Temperature
(D)
Absorptance A
Transmittance diffuse T_{diff}
Transmittance unscattered T_{unsc}
Reflectance R
optical properties
temperature
168 °C
(C)
Reflectance R
Transmittance T
Absorptance A
40 °C
170 °C
195 °C
PA 2200
PA 2202
PA 3200 GF
PA 1101
(E)
Polyamide 12
λ = 1940 nm
d = 200 µm
Reflectance
Transmittance
Absorptance
Polymer at rooom temperature
Polymer still in powder form
Polymer molten
Polypropylene
λ = 1940 nm
d = 200 µm
Reflectance
Transmittance
Absorptance
Polymer at rooom temperature
Polymer still in powder form
Polymer molten

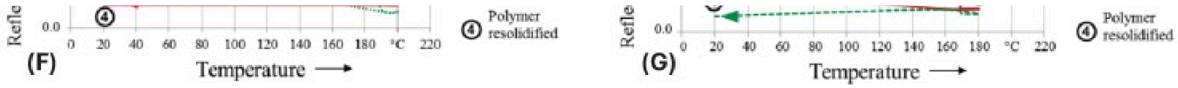


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Fig. 7. Graphical representation of microstructure induced thermo-optical property shifts of polymers as a function of temperature: (A/B) Reflectance – blue transmittance – orange and absorptance – grey of PA12 at 10.6µm for (A) 100µm and (B) 200µm layer thicknesses [74] (Permission No: 6123410942242). (C) Absorptance – red, transmittance (diffuse) – green, transmittance (unscattered) – blue, reflectance – grey of PA12 by temperature [77] (Permission No: 6123410570611). (D) Effect of layer thickness (200µm, 500µm, and 800µm) on optical properties (reflectance – grey, transmittance – red, absorptance – blue) of PA12 and PA11, (E) Optical properties (reflectance – grey, transmittance – red, absorptance – blue) of PA12, PA12CB, PA12GB and PA11, 200µm thick layers at 40°C and the material-specific preheating temperature [71] (Open access article CC-BY4.0). (F and G) Optical properties (reflectance – red, transmittance – blue, absorptance – green) of PP and PA12 powder by temperature at 1.94µm [78] (Permission No: 6123410771107). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Heinl et al. [74] found PA12 absorptance to drop sharply at melt onset, attributed to reduced optical light refraction as crystallites melt. Absorptance remained lower in the recrystallised material as the solidified material oriented more consistently versus the pre-sintered powder, increasing transmittance. PA12 across all three studies reported large absorptivity shifts, ranging from a 20% to almost 50% decrease before and after sintering. As the reflectivity remained the same, it should be apparent that the transmittance increased accordingly, making the loss of absorptivity less pronounced for thicker layers shown in Fig. 7 (D). Fig. 7 (C) shows the diffuse versus un-scattered transmittance of light, while (E) shows the difference between the lowest measured temperature (40°C), and the powder preheat temperature determined by DSC testing.

Notably in Fig. 7 (E), it is apparent that the ambient temperature and pre-heat temperature require separate calibration for regular PA11, PA12, and PA12GB, however, the PA12 with CB does not show this difference. Schuffenhauer et al. [71] indicates the CB suppresses the effect of increased transmittance in the PA12 powder, which is expected to hold true as long as no phase change occurs in the powder. This indicates the emissivity of CB filled powders may be easier to determine compared to those without, as the mechanism behind absorptivity and emissivity differs from unfilled powders. This was observed by [79] as decreasing diffuse reflectance at all wavelengths as the percentage of CB additive increased. CB-coated powders absorb a greater energy fraction at all wavelengths, resulting in higher sintering temperatures. Fig. 7 (F and G) shows temperature and phase-dependent emissivity shifts occur in other polymers, specifically comparing PA12 and PP [78]. It should be noted the wavelength of light used is 1.94 μm as opposed to the 10.6 μm wavelength characteristic of CO_2 lasers. Much like the PA12, the absorptance of PP decreases as the temperature increases, before rapidly decreasing at melting. Unlike PA12, the reflectance near mirrors the absorptivity in both magnitude and change. Upon solidification, the emissivity remains nearly unchanged from melting. The reflectance for both PA12 and PP are much higher at this wavelength.

These changes highlight the importance of emissivity in thermal imaging, as thermal imagers require calibration to the specific emissivity of the target object. Post-correction of the collected data is likely required if absolute temperature of the melt pool is required. Otherwise, it is suggested the user only consider the morphological distribution of values without much consideration of the reported temperature.

4.3. Thermal sensitivity

The most common figure of merit for thermal sensitivity is the Noise Equivalent Temperature Delta (NETD), infrequently referred to as NEDT or NEΔT [80], [81]. Poor thermal sensitivity yields noisy images, introducing uncertainty into the resultant image. This is separate from frame rate and resolution related blurring, higher NETD values can improve contrast in the image through reduced noise. NETD refers to the smallest temperature difference between pixels an imager can resolve, specifically the temperature difference in millikelvin between adjacent detectors required to equal the total noise of the camera (Signal=Noise, SNR=1). Other figures of merit exist, however are less broadly useful unlike NETD. Sensor calibration varies between manufacturers and thus recommendation for this varies. More general theory behind calculation of NETD is covered by Budzier and Gerlach [81], as does Miller [82].

Briefly, it is important to note NETD depends on the temperature of the measured object, the reference temperature is required for comparison [81]. The F-number of the applied optics also impacts the calculation, F/1.0 is standard [81]. Large variations in the temperature at which the test is conducted at will impact the comparability of the test.

4.4. Sensor types

As discussed earlier, selection of a thermal imager is highly dependent upon the user's process and data requirements. The two bands of interest in polymeric PBF are MWIR (~3.0–5.0 μm) and LWIR (7.0–14 μm). Due to higher temperatures yielding shorter wavelength emissions, Near-Infrared (NIR) and Short-Wave Infrared (SWIR) sensors perform poorly due to low sensitivity, with their lowest measurement temperature often exceeding the temperatures at which most polymers degrade. Of the relevant imagers, almost all belong to the radiation detector (bolometer), or photonic/quantum detector families [80], [82]. Bolometer type radiation detectors are typically cheaper due to photonic sensors requiring costly cryogenic cooling (e.g. 70–80 K [82], [83]), typically utilising closed-cycle Stirling cryocoolers (Stirling heat cycle cooler). This comes at the expense of bolometers possessing lower thermal sensitivity.

4.4.1. Bolometer detectors

Radiation detectors, typically bolometric, rely upon the temperature coefficient of resistance (TCR) of the detector to determine the incident thermal energy. Put simply, the bolometer film's resistance changes when it heats up from absorbed incident infrared radiation [82], [84]. Microbolometers typically consist of a suspended Vanadium Oxide

(VOx) or Amorphous Silicon (α-Si) film [85] connected to a read out integrated circuit (ROIC) [86]. This is shown in Fig. 8 (A and C), noting the suspended bolometer film is designed to reduce thermal conduction to the ROIC. VOx films are widely used due to favourable cost and performance characteristics [85]. Bolometer type sensors do not require cooling, unlike photonic type sensors [82], making them much smaller and more readily retrofitted onto existing OEM machines. The uncooled microbolometers are typically VOx, however some manufacturers use α-Si. Older models potentially utilise Barium Strontium Titanate detectors (BST), though this is now uncommon.

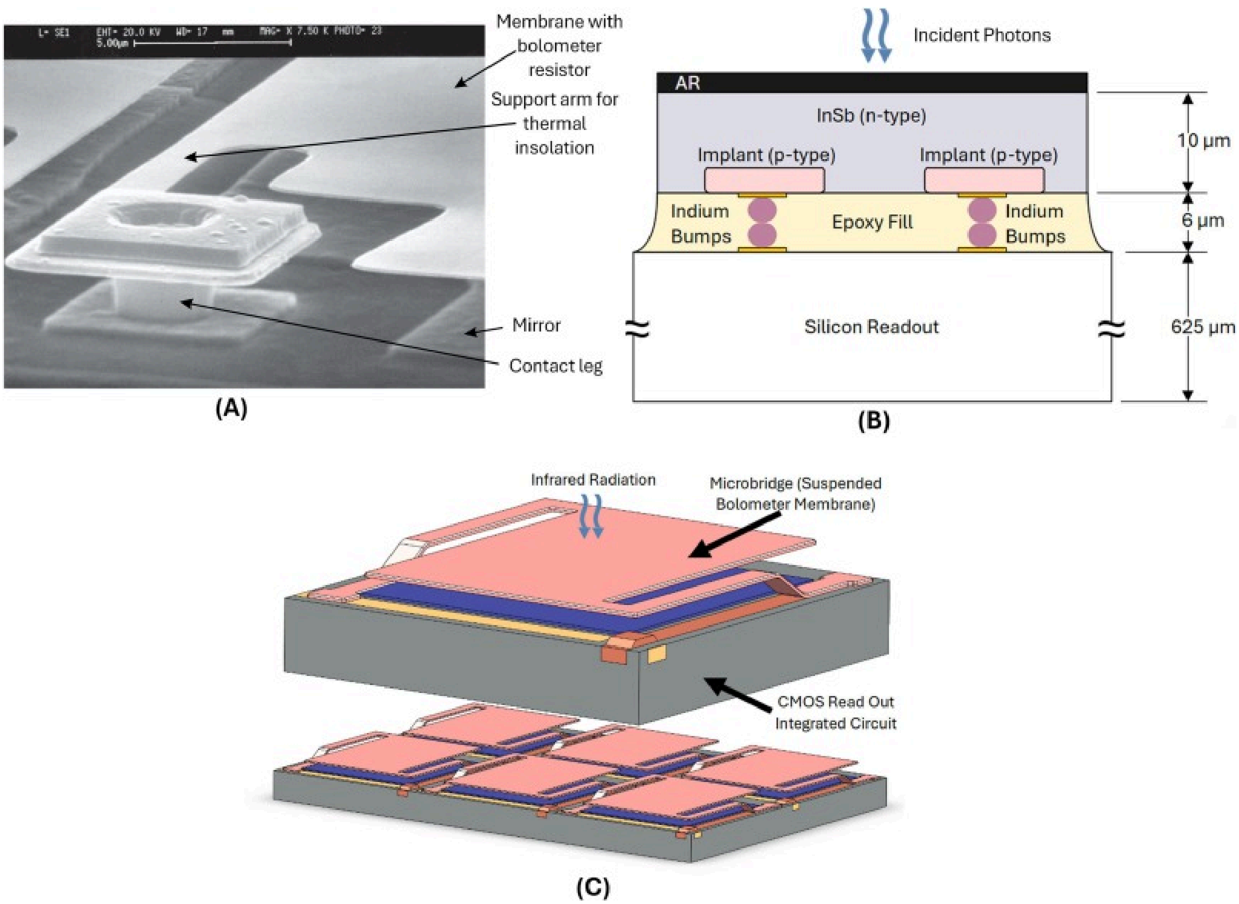

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Fig. 8. Bolometer and photodetector sensor diagrams. (A) Microbolometer pixel structure, pixel pitch 50 μm; bridge height 2.5 μm (provided to [92] by ULIS, Veurey Voroize, France) (Permission No: 6123490610333). (B) Example of material stack-up in flipped chip InSb detector (adapted from [94]). (C) Representative microbolometer cell showing suspended microbridge and bonded CMOS read out (adapted from [86]).

For microbolometers, NETD values around 20 mK have been achievable since the late 1990s [87], noting a given process typically exhibits a bell-curve distribution of achievable NETD values. Mean NETD values in the mid-30 mK to 50 mK have also been demonstrated for several years now [88], [89], [90], several models meeting this are readily available across a variety of price points. They offer exciting opportunities to the researcher now at much lower cost than until recently, and their absence from the reviewed literature is indicative of opportunity where absolute temperature is not required.

4.4.2. Photonic detectors

Of the photonic detectors, photovoltaic types are more suited to imaging applications due to lower thermal losses [82]. Quantum well infrared photodetectors and photoconductive sensors are included in this family [80], [82]. Photovoltaic sensors rely on the photovoltaic effect, measuring the voltage difference created by incident photons generation of electron and electron hole pairs [82]. Typical compositions noted in polymeric PBF applications are limited to Indium Antimonide (InSb) and Mercury Cadmium Telluride (HgCdTe/MCT), which may be attributed to the favourable spectral window including but not limited exactly to the 3.0–5.0 μm range [91], as well as the economies of scale supporting their usage. Photonic detectors can be thought of as the traditional, high-resolution, high-sensitivity, and high-frame-rate option, coming with accordingly high prices. Fig. 8 (A and B) show basic stack-ups for InSb flip-chip sensors, which fabricate the detector and ROIC separately, before bonding with indium bump bond connections [92], [93].

4.5. Sample rate

Sample rate, or the frame rate of a sensor, is an important metric depending upon the timescale of the monitoring target phenomena. Determination of this differs depending upon sensor type.

4.5.1. Microbolometer thermal time constant

Microbolometers, unlike photon detectors, measure the resistivity of each pixel's bolometer membrane [84]. The response time is largely dictated by the thermal time constant $\tau = \frac{C}{G}$, where C is the heat capacity of the bolometer pixel and G is the thermal conductivity between the bolometer and its surrounding [84]. Typical thermal time constants are around 7–15 ms [83], [95]. Frame rates are typically around or below 60 Hz, although up to 100 Hz has been demonstrated with a NETD of less than 120 mK [96]. The overall time constant is limited by thermal time constant, ROIC and the readout mode [96].

4.5.2. Photon detector integration time

Photon detectors collect energy during the desired integration time [82]. Longer integration times improves sensitivity, as it increases the signal-to-noise ratio, noting this will decrease the overall frame rate of the output signal. Accordingly, lower integration times may be set to increase frame rate, leaving the ROIC and input/output electronics as the main bottleneck. This is beneficial to the user who may use it to capture both whole-of-bed thermographs, then switch to higher-frame rate, lower resolution modes to further investigate regions of interest.

4.6. Considerations during usage

This section discusses application specific considerations for implementations of thermal monitoring to existing or conceptual machines. Discussion of resolution and frame rate suitability to requirements, as well as mounting considerations are presented.

4.6.1. Data management and machine learning

High-resolution and high-frame-rate imaging whether in polymer or metal AM generates a large amount of data, requiring significant computational power and time to process [97]. Machine Learning (ML) more generally requires large and diverse datasets to train; however, the resultant model can process large datasets efficiently [97]. Additionally, data quality is important due to the need for prior training of machine learning models, which can also influence the possibility of overfitting or underfitting of machine learning predictions. Furthermore, ML can enhance efficiency in prediction of both polymeric and metallic manufactured part properties, with the ability to evaluate a high quantity of data in real time. As a technique, ML enables manufacture of complex geometries with higher efficiency compared to traditional manufacturing methods. This leads to less material waste, cost, and time. One area of particular interest is ML for processing of thermal imager data. Integrating machine learning processing of in-situ thermal monitoring data can be utilised to improve build parameters, yielding greater control of part strength, crystallinity, porosity, morphology and minimisation of defects in printed parts [98], [99], [100], [101], [102].

4.6.2. Temporal and spatial requirements of polymeric PBF

Sample rate of a given camera is dependent upon numerous factors, which varies for bolometer-based sensors, and photonic/quantum sensors. Despite this, Fig. 9 [17], [39] shows the higher viscosity of polymer melt versus metals results in much larger time-scales (around 10–20s) for particle sintering and densification of layers, indicating imaging of solidification and densification of the polymer melt can be achieved with relatively slow cameras. Pulsed-wave and continuous-wave laser sintering of PA12 and PEKK have been shown to have similar resultant temperatures, however this only held true for scanning speeds sufficiently low that the laser spot pulses overlapped [17].

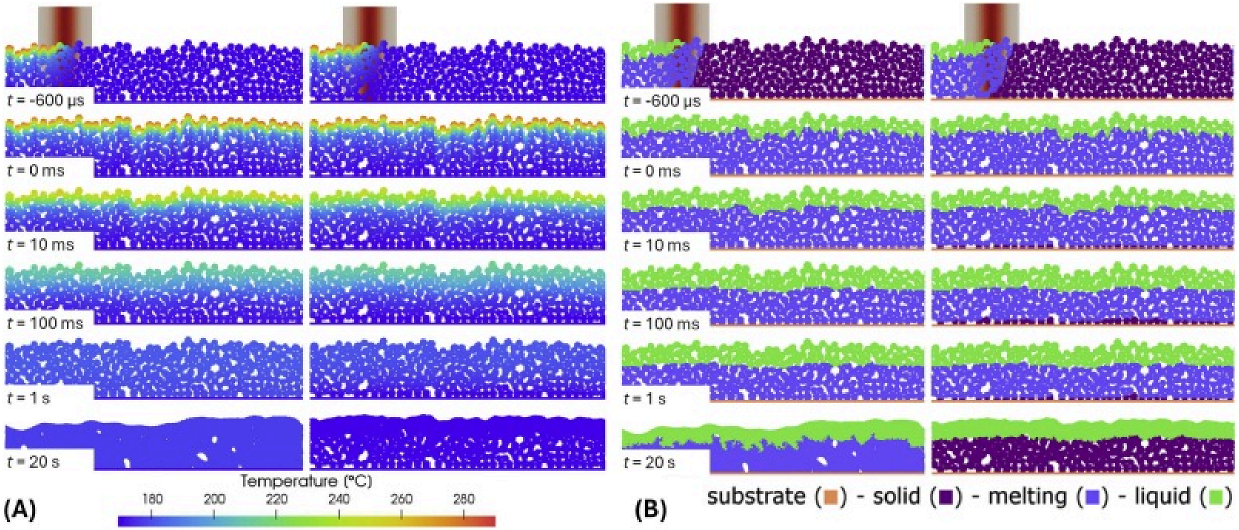

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Fig. 9. Simulated sintering time scales with adiabatic and with cooling boundary conditions: (A) by temperature (left adiabatic, right cooling), (B) by phase (left adiabatic, right cooling) [39] (open-access article CCBY4.0).

Taylor [37] found a wide-area, higher resolution thermal imager at 30Hz better able to predict fracture location in tensile test specimens versus a bore-sighted, lower-resolution subframe at 2.24kHz. This type of setup is illustrated in Fig. 10 (B), where a coaxial FLIR SC8420 MWIR, wide-area FLIR A6701 MWIR, and visual camera are mounted simultaneously. Thermal fields immediately prior to sintering were more highly correlated to mechanical properties than thermographs taken directly after sintering of the layer, this was attributed to post-sintered images introducing error through varying phase states and thus emissivity. It is thus suggested high-speed monitoring is only necessary for more specific monitoring such as melt pool analysis.

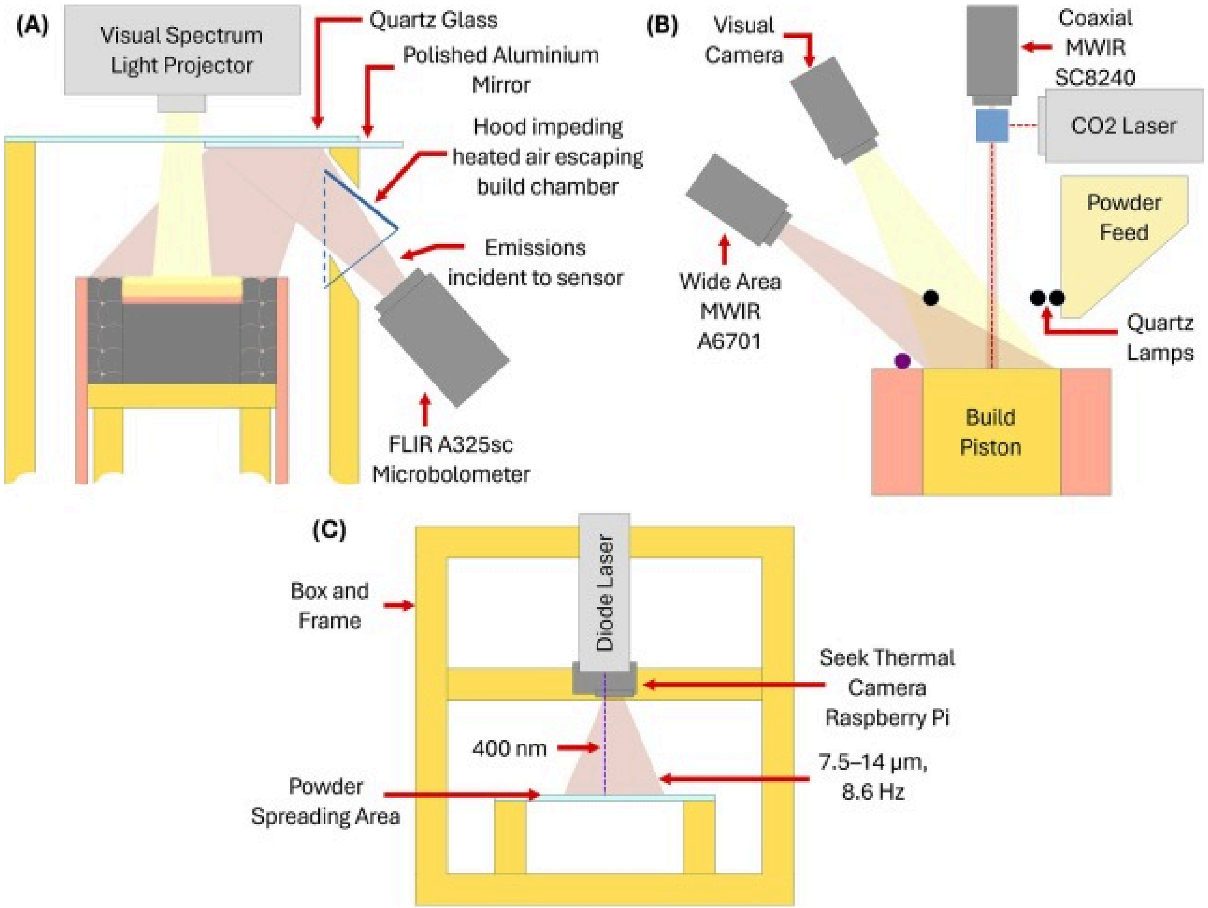

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Fig. 10. Thermal imager installation on custom research polymeric PBF machines: (A) Lower cost microbolometer setup utilising mirror to view powder bed (Adapted from [48]). (B) Three-viewport setup of the University of Texas at Austin's SLS machine with bore-sighted MWIR and wide-area MWIR (Adapted from [37]). (C) Low-cost consumer microbolometer setup on custom SLS testbed simplified annotated view (Adapted from [105]).

4.6.3. Sensor mounting

The angle between the print bed plane and thermal imager axis is important, the error increasing for rising angles. Fig. 10 (C) illustrates a near coaxial, centralised mounting of a low-cost consumer Seek Thermal microbolometer, the mounting of which minimises optical distortion. Mounting angle however can usually be considered negligible on temperature accuracy below 30° off axis, or per manufacturer recommendation [103]. Measurement error from vertical viewing angles greater than 45° is accentuated, while angles in excess of 75° are impractical due to being near parallel to the emission plane [104]. Sensitivity to viewing angle is worse for smaller fields of view [104]. For certain geometries and parameters, sintering exposed layers can result in surface sinking relative to the unsintered powder bed, potentially resulting in a change in camera focal distance, and thus reduced detection of thermal radiation [67]. Retrofitting of microbolometer type sensors to machines may require modification as they are unable to view radiation through typical visual-light transparent windows, Gardiner [48] utilised a polished aluminium mirror similar to Fig. 10 (A) to view the reflected infrared radiation.

4.6.4. Thermal considerations

Build chamber temperatures may exceed the OEM safe operation temperature of both cooled and uncooled sensors. Use of a low-cost FLIR Lepton 3.5 in an air-cooled PA12 housing was demonstrated by [42], who noted uncooled infrared windows may suffer absorption coefficient and infrared radiation transmission shifts. The significance of this is uncertain, overall viewport assembly heat loss for example was 2610 Wm^{-2} in the University of Texas at Austin LAMPS machine [32], although all three optical viewports were insulated. Radiative cooling was indicated to dominate heat transfer in the build chamber with convection far less dominant, making the Boussinesq approximation appropriate [38], however it should be noted some sensors may have their view occluded by certain gases [82], potentially requiring spectral filtering to avoid imaging the convective heat transfer. Soundararajan et al. [105] demonstrates usage of a small single-board computer as a datalogger for a low-cost microbolometer designed to connect to mobile phones as illustrated in Fig. 10 (C). This approach may be considered for internal to machine applications where the ambient temperature is sufficiently low, usage of temperature sensitive labels may be considered to determine this.

4.6.5. Measurement methodology and associated inaccuracies

The projection of thermal imager pixels onto the targeted scene is referred to as the Instantaneous Field of View (IFOV) [106], sometimes erroneously referred to as the resolution of the thermal imager [107]. Limitations of the optical system's diffraction limit, aberration, scatter, diffusion of charge in the sensor and other factors contribute to the final spatial transfer function of the instrument and blurring between neighbouring pixels [107]. Another effect to be wary of is the keystone effect, which occurs when the imager is not placed perpendicular to the viewed image, resulting in a skewed image. Graphically, this can be compensated for by the datalogger, or later with software functions such as the *imtransform* function built into MATLAB.

Contamination of optical windows in the build chamber is prevalent (see Fig. 11 B and C) [42], [108], often mitigated with a gas curtain projected in-front of the viewport [109], though other approaches such as heating of the window to prevent condensation of vapour has been successfully demonstrated [108]. Failure to address this impacts laser efficiency by 20–25% by reducing transmission [59], which will likewise impact thermal monitoring. Filtering may be necessary to avoid damage to sensors, evaluating material absorptivity and reflectance, and process type emissions and sensor spectral sensitivity is important. Use of protective filters may be necessary when the powder has high reflectance, and there is insufficient attenuation of the laser energy between the powder and the monitoring system [74]. This can be avoided by using a thermal imager with a wavelength sensitivity separate from the input wavelength, for example [37] uses MWIR imagers with a sensitivity from 3.0 to 5.0 μm, with a typical CO_2 sintering laser operating at 10.6 μm. LWIR sensors with a sensitivity around 7.0–14 μm are more appropriate for HSS and MJF as the fusing lamp should be a lower wavelength outside the spectral sensitivity of the imager.

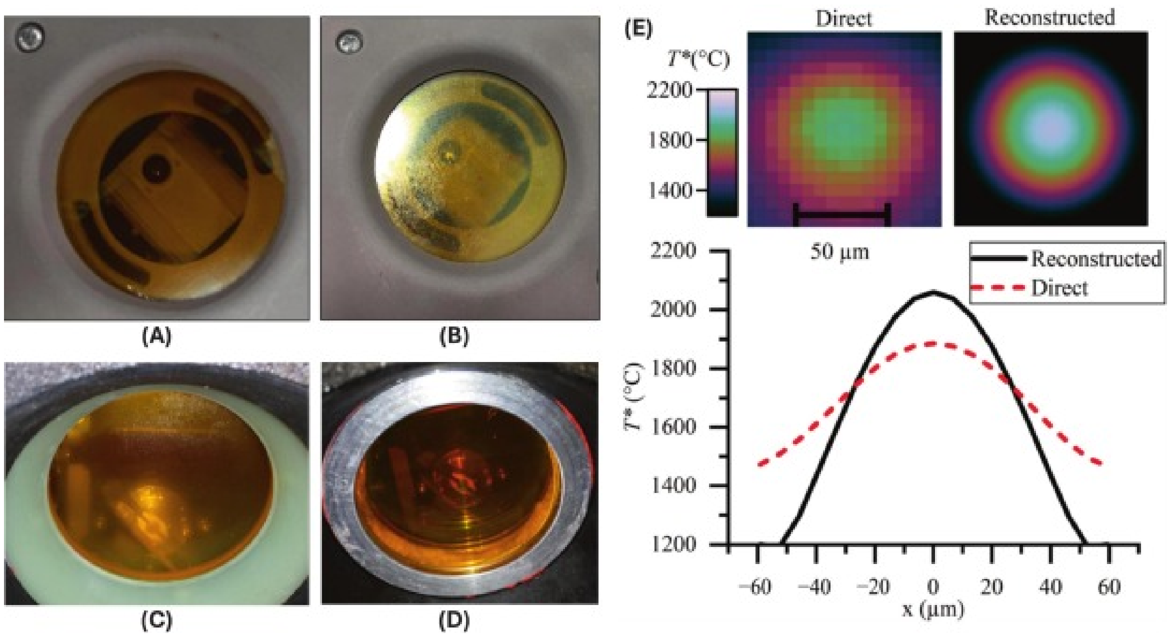

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Fig. 11. Optical challenges in thermal monitoring of polymeric PBF: (A) Protective ZnSe optical window for thermal imager, (B) surface contamination from print [42] (Permission from author). (C) surface contamination on window, (D) modified heated window to prevent condensation [108] (Re-use permitted with citation). (E) Thermal image reconstruction error and modified reconstruction demonstrated [107] (open-access article CCBY4.0).

4.6.6. Band notation ambiguity

Numerous popular subdivisions and acronyms for the infrared spectrum exist [81], [82], [91], [110], including DIN [111], CIE [112] and ISO [113] standards. Generally categorised as between 780 nm (0.78 µm) and 1 mm (1000 µm), the ranges individuals consider useful are highly dependent upon the field in question. A non-comprehensive but hopefully useful visualisation of commonly encountered (and cited) subdivisions in polymer AM is presented in Fig. 12. A brief summary will be presented for each scheme presented.

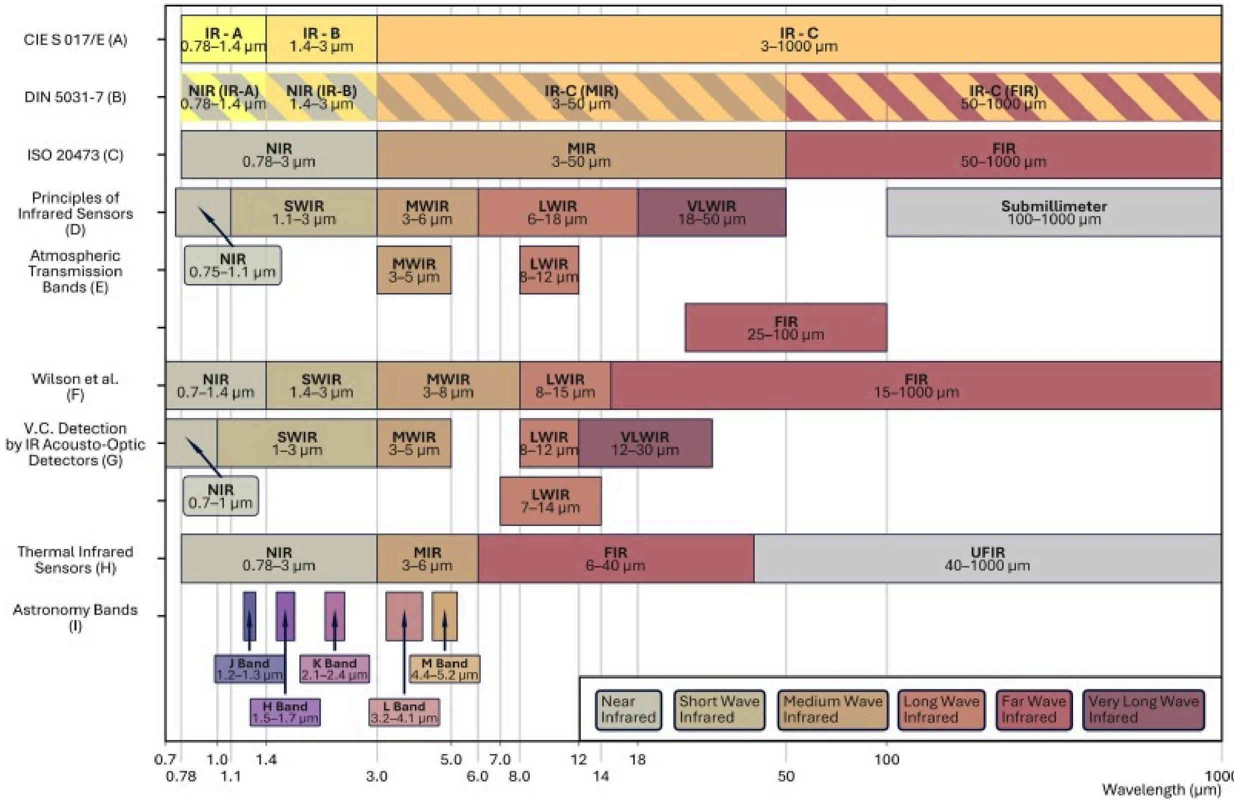

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Fig. 12. Commonly used Infrared Radiation Band acronyms and associated wavelengths in polymeric PBF literature, (A/B/C) International Standards [111], [112], [113]. (D/E) Common first citation for atmospheric transmission bands [82]. (F) IEEE sensors publication on usage in ML [110]. (G) Wavelength acronyms by common sensor wavelength sensitivity [91]. (H) Another common first citation for band acronyms [92]. (I) Bands used for astronomical purposes however infrequently of interest [82], [92].

The CIE standard [112] is the oldest standard circa 1938, denoting ranges as IR-A, B and C. This partially overlaps with ISO 20473 [113] and DIN 5031–7 [111], who alternatively use near-infrared (NIR), medium-infrared (MIR), and far-infrared (FIR) abbreviations. None of these ranges are of particular interest to thermal monitoring of polymers as the regions of interest, 3.0–14 μm, all fall under the collective region of IR-C or MIR.

Referencing older, frequently encountered schema, Miller's *Principles of Infrared Technology* [82] was the oldest reference noted in English speaking polymeric PBF literature. Budzier and Gerlach's *Thermal Infrared Sensors* [92] is similarly popular and has a similar range, though less useful due to FIR encompassing a large range. D'Amico et al. [91] is another popular scheme, presenting Miller's definitions modified slightly to represent the actual range achievable by typical sensor technologies.

Due to the excessive variation in spectral range encompassed by terms such as MWIR and LWIR, it is suggested that users refer to the manufacturer's specification and clearly specify the spectral range measured. Atmospheric transmission bands MWIR (3.0–5.0 μm) and LWIR (8.0–12 μm) are the most generally useful two classifications for polymeric PBF. They are based on usable ranges for heat-seeking missiles, the 5.0–8.0 μm range unusable due to atmospheric attenuation of those wavelengths [81], [114], [115], MWIR and LWIR ranges incidentally aligning well with both commercial sensor ranges and the required range for polymeric PBF monitoring.

4.7. Overview of identified thermal imaging sensor models

The evaluated literature and their associated findings are presented in Appendix Table A. A summary of unique models identified, and their specifications are listed in Table 2. Two major sensor types were identified – uncooled LWIR (7.0–14 μm) microbolometers offering lower-cost, lower-frame rate thermal imaging in a smaller package, and cooled MWIR (3.0–5.0 μm) InSb detectors. Though uncommon, cooled HgCdTe sensors were also used.

Table 2. Unique sensor models noted in literature. Sensor parameters largely extracted from manufacturer datasheets.

Model	Manufacturer	Resolution	Sensor family	Sensor	Wavelength	Frame rate	NE
SC8240 \ SC8243	FLIR	1024×1024	Photovoltaic	InSb	3.0–5.0 µm	125 Hz	<2
A6701	FLIR	640×512	Photovoltaic	InSb	3.0–5.0 µm	60 Hz	<2
A325	FLIR	320×240	Microbolometer	VOx	7.5–13 µm	60 Hz	50
A600 series	FLIR	640×480	Microbolometer	VOx	7.5–14 µm	5 Hz	50
Lepton 3.5	FLIR	160×120	Microbolometer	VOx	8.0–14 µm	8.6 Hz	<5
SC4000	FLIR	320×240	Photovoltaic	InSb	1.5–5.0 µm	420 Hz	<2
SC7000 MWIR \ SC7600	FLIR	640×512	Photovoltaic	InSb	1.5–5.1 µm	100 Hz	<2
T420	FLIR	320×240	Microbolometer	Likely VOx[a]	7.5–13 µm	60 Hz	<4
X6901	FLIR	640×512	Photovoltaic	InSb	3.0–5.0 µm	1004 Hz	<2

Model	Manufacturer	Resolution	Sensor family	Sensor	Wavelength	Frame rate	NE
Ti400	Fluke	320×240	Microbolometer	Unspecified	7.5–14 µm	60 Hz	50
ImageIR 5300 \ ImageIR 5385	Infratec	320×256	Photovoltaic	HgCdTe	3.7–4.8 µm	481 Hz	<1
ImageIR 8300	Infratec	640×612	Photovoltaic	InSb/HgCdTe	1.5–5.5/5.7 µm	151/205 Hz	<2 ml
Jade III	Infratec	320×240	Photovoltaic	InSb/HgCdTe	3–5.2 µm	170 Hz	<2
Velox 1310k SM	IRCAM	1280×1024	Photovoltaic	InSb	1.5–5.5 µm	180 Hz	<2
TIM 400	Micro Epsilon	382×288	Microbolometer	Unspecified	7.5–13 µm	80 Hz	80
TIM 640	Micro Epsilon	640×480	Microbolometer	Unspecified	7.5–13 µm	32 Hz	75
P640i	Optris	640×480	Microbolometer	Unspecified	8.0–14 µm	32 Hz	40
Xi400	Optris	382×288	Microbolometer	Unspecified	8.0–14 µm	80 Hz	50
Compact	Seek Thermal	206×156	Microbolometer	VOx	7.5–14 µm	8.6 Hz	Ur

Model	Manufacturer	Resolution	Sensor family	Sensor	Wavelength	Frame rate	NE
890–2	Testo	640×480	Microbolometer	α-Si	7.5–14 µm	33 Hz	<4
A310	FLIR	320×240	Microbolometer	VOx	7.5–13 µm	30 Hz	<5
E40	FLIR	160×120	Microbolometer	VOx	7.5–13 µm	60 Hz	<7
348 A	Fotric	640×480	Microbolometer	Unspecified	7–14 µm	30 Hz	<3
DT-980	Huashengchang Technology Co.	80×80	Microbolometer	Unspecified	8.0–14 µm	50 Hz	<1
875i	Testo	160×120	Microbolometer	Unspecified	7.5–14 µm	33 Hz	<5

a

FLIR Bolometers are likely all Vanadium Oxide based.

High-frame rate, sensitivity, and resolution sensors are dominated by InSb sensors, shown in Fig. 13. The most frequently noted sensor was the IRCAM Velox 1310 k SM, however shared co-authorship indicated a common research group was likely utilising the same research setup, indicating the broad applicable usefulness of robust thermal monitoring setups. Of note, Fig. 13 and Fig. 14 only list the full-frame resolution frame rate, with InSb detectors usually capable of sub-frame windowing modes at much faster frame rates.

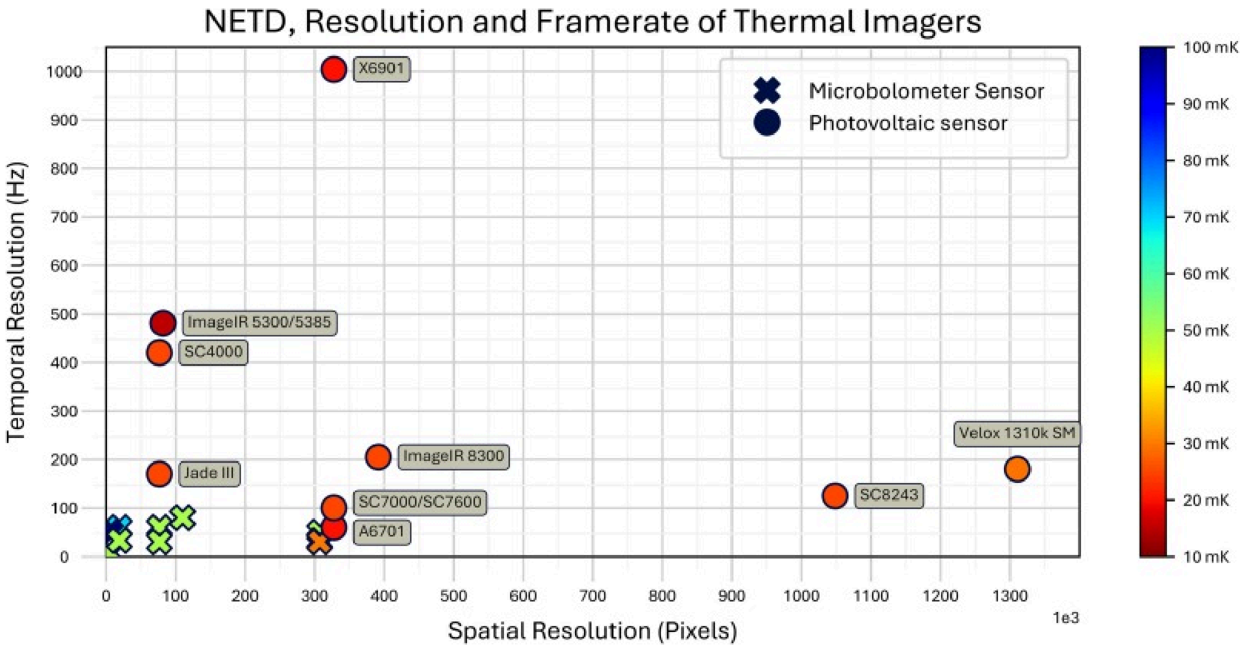

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Fig. 13. NETD, Resolution, Frame rate and sensor type of cameras identified in Polymeric PBF publications.

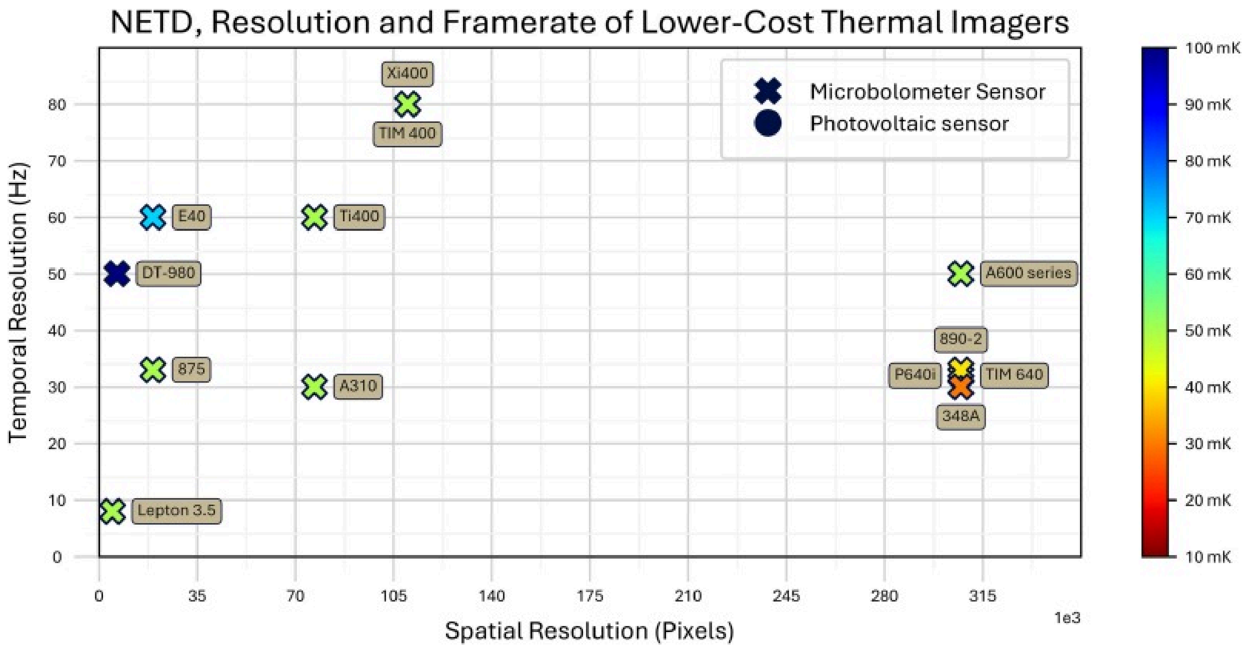

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Fig. 14. NETD, Resolution and Frame rate of cameras identified in Polymeric PBF publications, low frame rate, and resolution models.

InSb detectors are expensive, however the excellent resolution, frame rate, and sensitivity lend themselves well to imaging transient behaviour in SLS. The MWIR equivalent to the FLIR X6901sc, the X6900sc has a maximum frame rate of 29 kHz at a 640×4 sub-frame [116], [117]. The MCT Infratec ImageIR 5300 can achieve up to 105 kHz in sub-frame mode [118].

Microbolometer sensors offer lower resolution and lower frame rates typically between 30 and 60 Hz for scientific imagers. Low frame rate is an intrinsic limitation of the operating principle; however, the sensor does not require cooling and thus presents a smaller overall package. The microbolometers indicated were typically retrofitted to custom machines or used handheld imagers. The thermal sensitivity is typically low, however the cluster of Testo 890–2, Optris P640i, Micro-Epsilon TIM 640 and Fotric 348A shown in Fig. 14 have a high likelihood of sharing the same sensor. All four sensors showed much better thermal sensitivity than other microbolometers listed. Note Fig. 14 merely shows the cluster of lower resolution and frame rate sensors, inclusion of solely microbolometers was a result of the input dataset and not intentional on the part of the author. This does, however, highlight the difference in typical utilisation and capabilities between microbolometer and photovoltaic type sensors.

5. Other in-situ techniques

Other forms of contact and non-contact thermal monitoring are used in absence or in conjunction with thermal imagers. Combined sensing approaches are typical for research machines, for example the University of Texas at Austin LAMPS machine used over forty strip heaters with independent thermocouples alongside a FLIR A6701sc MWIR for build surface heater monitoring, and a FLIR SC8420 MWIR coaxial to the sintering laser [40], though no direct evaluation of the strip heater control was presented. Discussed earlier, thermal imagers have limitations capturing the true temperature of an image, even with black-body references [70]. Usage of thermocouples as reference [33], [159], [160], [161] is one method of addressing this, though determining melt pool temperature is not feasible with this method. Similarly, as a surface monitoring technique, measuring internal powder cake temperatures is impossible.

More novel approaches discussed include evaluating thermal induced defects with laser profilometry [14], [26], and internal powder cake temperature measurements with microwave computer tomography [162].

5.1. Thermocouples and thermometers

Thermocouples and thermometers are ubiquitous and relatively low cost, presenting an accessible method of thermal monitoring. Key considerations for implementation and discussion of their limitations are presented.

5.1.1. Internal powder cake measurements

Kiani et al. [33] attached five thermocouples to the build plate of a custom SLS test machine in Fig. 15 (A). Arranged in a cross pattern with 10 mm between probes—preheat, printing, and cooling temperatures were monitored for PA12 tensile test specimens-built edgewise and flat-wise approximately 500 µm from the thermocouples. Minimal temperature difference between the five thermocouples was observed as shown by Fig. 15 (B). The initial proposed hypothesis assumed different cooldown rates would influence the thermal history of the produced parts, yielding different degrees of crystallisation. Despite distinctly differing cooling rates (sintering temperature to cooldown in approximately 50, 100 and 150 min), minimal crystallinity variation was observed using DSC, with the maximum variation 3.8%.

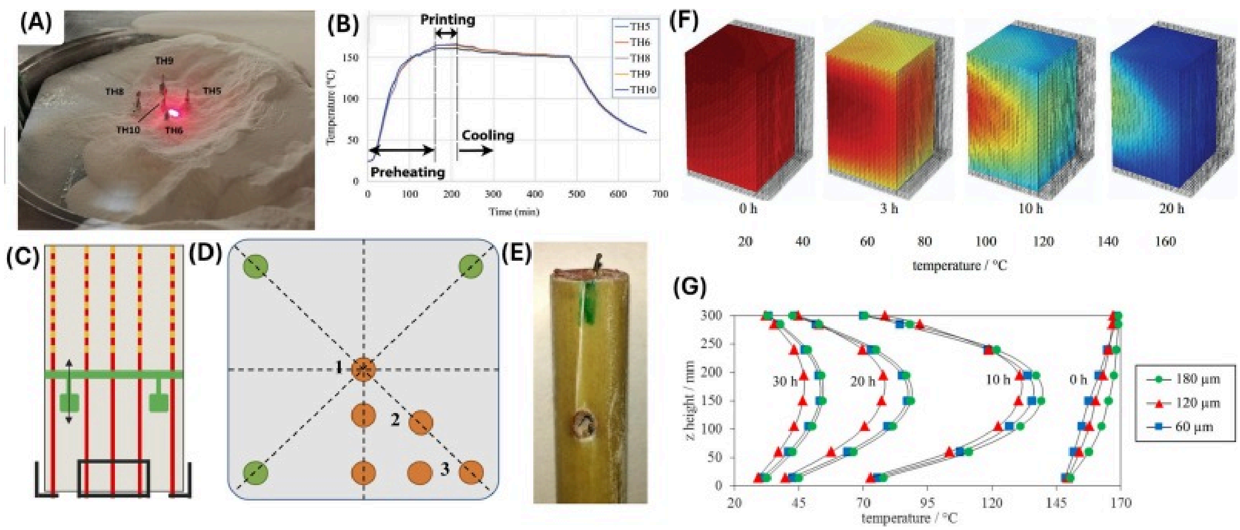

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Fig. 15. Powder cake thermal monitoring with thermocouples: (A) Thermocouples embedded into powder cake, (B) thermal data [33] (Permission No: 6123600219562). (C) Thermocouple sensor bar front view, (D) top view, (E) sensor bar design, (F) 3D

visualisation of powder cooldown from measured and interpolated values, (G) Cooling profile in *Z*-direction for different layer thicknesses [161] (Permission No: 1712738-1).

Josupeit and Hans [161] modified a EOSINT P395 SLS printer with nine sensor bars shown in Fig. 15 (C, D and E) to evaluate the temperature history in the build powder cake. Holes were drilled into the build platform allowing the sensors to be embedded within the build. Each sensor bar had eight thermocouples, though the build area was measured by a cluster of 48 sensors. Datalogging was done with a Delphin Technology Expert Key 200L. Consistent build heights of 300 mm were conducted at 60, 120 and 180 μm layer thicknesses using 100% recycled EOS PA2200 PA12 powder. The resultant thermal data was interpolated and plotted in MATLAB, shown in Fig. 15 (F and G). The authors noted clear elliptical isothermal shells as the outer powder cake lost heat to the ambient environment, while the inner powder remained insulated. Subsequent work by Josupeit and Hans [160] used the same setup to create a finite element thermal model of the powder cake temperature. The front, side, and top of the powder cake was additionally derived using PT100 resistance temperature sensors. The thermocouple sensor bars were reported to have an average standard deviation of 1.6 K and Root Mean Square Difference (RMSD) of 2.5 K between experiments with the same build conditions. This demonstrates the potential of well-designed thermocouple placement in validation of thermal simulations, though it should be noted substantial modification of the base machine was required. One notable flaw identified by the authors was discontinuity in the temperature fields introduced by cracks in the powder bed which was not modelled.

Diller et al. [159] measured the powder cake temperature near the built part on a 3D Systems Sinterstation 2500 SLS. Two Omega RTD-4-F3105–36-T, 100 Ω 4-wire resistive temperature detectors (RTD) were used to measure the centre and corner of the build piston temperature, mounted 25 mm from the build piston face with spacers made from cork. Data logging was conducted with a National Instruments NI 9217 data acquisition system; the ADC resolution was 24-bits. A FLIR A325 thermal imager was used to measure the top-surface temperature. A clear thermal gradient between the colder corner and warmer centre was noted with the corner probe decreasing in temperature faster than the centre probe, indicating the edge of the powder cylinder cooled faster than the centre as one would expect. This data was correlated against 1D and 2D thermal simulations, with the experimental data largely matching the simulation, with the exception of the experimental data possessing a 10 °C lower peak temperature, and slightly faster cooldown. This was attributed to the thermal conductivity of the RTD's wires being higher than the powder, providing a more preferential path for heat transfer to occur. This highlights the difficulty introduced using contact sensing, the placement is difficult and may alter the thermal fields one wishes to measure.

5.1.2. Thermal imager emissivity calibration with contact probe

Kubiak et al. [163] used a K-type thermocouple with a Omega 871 A digital thermometer to collect ambient temperature for calibrating a FLIR A325 thermal imager. The thermal imager captured the upper surface of dropped powder on a plate, with the lower surface temperature measured with three K-type thermocouples and an Omega OMB-DAQ-2416 DAQ. Accordingly, the powder emissivity was calculated based off the two sets of readings. Measuring emissivity of the powder in the machine is important due to the previously mentioned emissivity shifts during heating. This is exacerbated by the particulate, with the voids and holes also impacting emissivity. Numerical estimation of the emissivity was validated through the experimental setup; the thermal camera emissivity being adjusted to correlate with thermocouple readings. Similar to this, [105] used a Bosch MCP9808 digital temperature sensor to calibrate a Seek Compact 206×156 thermal camera. They evaluated the difference in temperature between the surface and build plate with a layer thickness of 1 mm. This technique of thermal imager in conjunction with contact thermal measurement is recommended to set the imager emissivity.

Thermocouples are both cheap, ubiquitous, and relatively accurate. When mounted by the OEM to the build piston, or heating elements, they can be used to determine environmental parameters accurately with minimal cost. Retrofitting thermocouples to machines adds a great deal of complexity which should be considered carefully, however.

Thermocouples and their wiring are typically more thermally conductive than the powder bed, when placed inside the powder cake care should be taken to consider this aspect. One key benefit they can be used for is calibration of thermal imagers, however placement of this will be highly machine dependent.

5.2. Pyrometers

Fibre optical heads of pyrometers can be more easily mounted coaxial to the laser optical head, simplifying packaging [70]. Typical pyrometer measurement in SLS shares the scan head with the sintering laser, Zander et al. [164] shown in Fig. 16 (A) demonstrates usage of a second galvanometer scan head which enables the pyrometer to vary the measurement spot in relation to the sintering laser. They demonstrated the feasibility of faster feedback with a spot sample rate of 50 kHz on a custom field programmable gate array controller. Subsequent investigations demonstrated both real-time feedback control of the laser to minimise warpage, as well as full-chamber temperature thermal field mapping (non-radiometric) enabled by the high sample rate of the pyrometer, scan head and field programmable gate array controller [29]. This approach is highly flexible once set up, as the pyrometer can image any region the user desires. Careful consideration of the viewed region and aliasing is important for this use case however, as unlike thermal imagers, the region of interest is no longer constrained to a fixed grid. OEM examples suitable for polymeric PBF monitoring applications such as the MWIR (2.2–6.0 μm) Optris CTLaser 4M have much lower frame rates (11 kHz) [165] than what was demonstrated Zander et al. Higher frame rate pyrometers are typically NIR optical type (Silicon detector, only measuring wavelengths below 1.1 μm) with minimum temperature thresholds in excess of 100 °C, unsuitable for polymers [166].

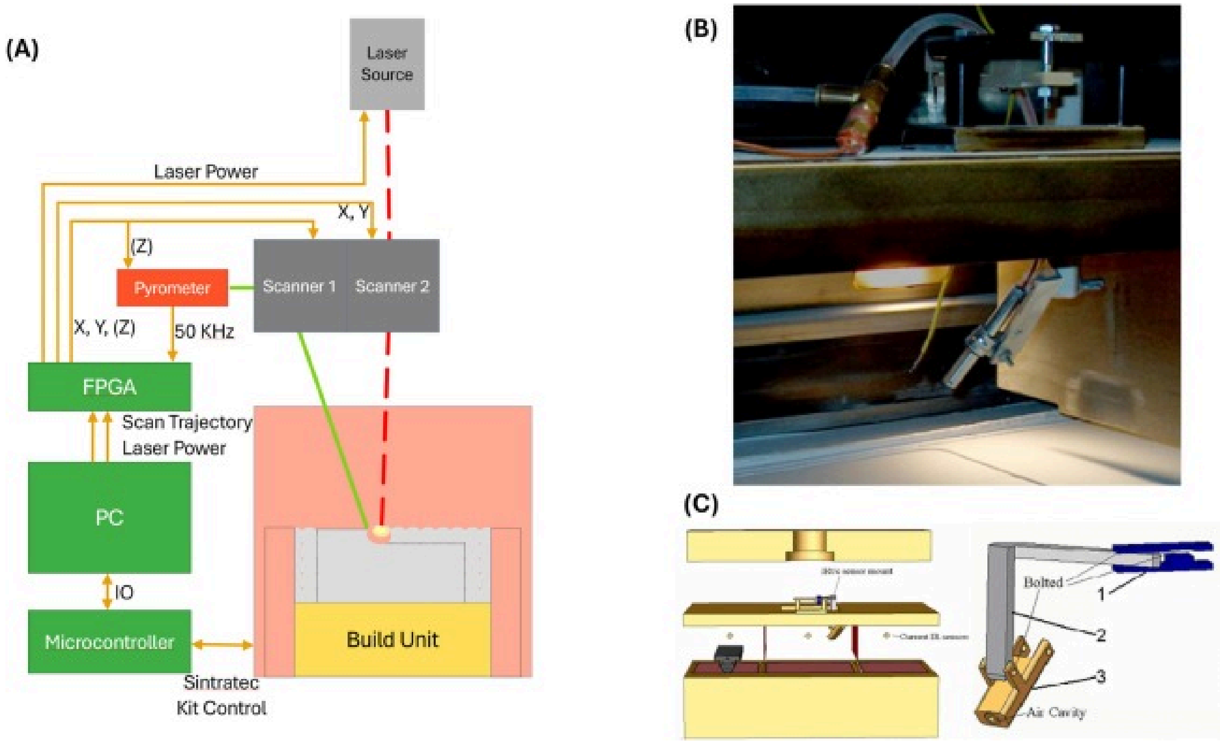

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Fig. 16. Pyrometer usage in polymeric PBF: (A) Independently scanned high frequency pyrometer, annotated componentry and signal diagram (adapted from [29]). (B) Retrofitted pyrometer in build chamber of a Sinterstation 2500Plus SLS, nitrogen cooled [167] (permission from author).

Kumpaty et al. [167] used an Exergen microIRt/c.4 infrared thermocouple for non-contact thermal monitoring. The selected sensor recommended air-cooling with a 17 L/min, 5 PSI air-line to enable operation up to 175 °C. Kumpaty et al. instead utilised a nitrogen feedline on the Sinterstation 2500plus SLS to cool the sensor, preventing addition of extra oxygen lines and introducing oxygen contamination. The selected sensor is reported to have a repeatability within 1%, and a maximum measurement temperature of 500 °C.

Initial tests yielded unexpected, declining temperatures which was the result of powder accumulation on the sensor over time. Comparison to the in-built machine sensors after cleaning the sensor was favourable to the modified sensor, with the higher sample rate capturing temperature fluctuations the OEM sensors could not. One notable limitation of this approach is the averaging behaviour of the infrared thermocouple's spot size, which limits the sensor's ability to precisely determine a region's temperature. It was noted the spot size was insufficiently small to capture the powder melting temperature. Consideration of how to keep the sensor clean is important to prevent changing transmission loss over time (Fig. 16 (B)).

Pyrometers are functionally cheaper, single point measuring thermal cameras. Many of the same considerations apply, however the cost is much cheaper, and the achievable frame rate is much higher. Typically utilised by OEM machines to monitor powder-bed temperature, the degree of access to machine sensors will vary by manufacturer. More niche applications may be achieved through usage of separate galvanometers as demonstrated by [164]. Due to the relatively low cost of quality microbolometers now, usage of pyrometers is increasingly less appealing due to the absence of any spatial information.

5.3. Novel approaches

Novel approaches to thermal monitoring in polymeric PBF are presented, noting the techniques may not be novel concepts more broadly in PBF. These techniques typically pose challenges in implementation such as usage of silicon CCDs, or are generally undeveloped areas of inquiry such as Microwave Tomography thermal monitoring.

5.3.1. Microwave tomography (MWT)

Sillani et al. [162] derived powder-cake temperature of a PA12 build on an EOS P1100 SLS, retrofitting a series of transmit/receive antennas to the build chamber where Fig. 17 (A) shows the CAD model highlighting salient changes, and Fig. 17 (B) shows the actual modified machine. Temperature dependent dielectric permittivity was derived from cylinder in an oven (diameter 180mm, height 30mm). 16 Transmit/Receive antennas were then used to reconstruct the dielectric characteristics of a printed cylindrical sample in the build chamber shown in Fig. 17. With a sample time of 90s, the technique currently has clear limitations in data acquisition time. Despite this, the thermal distribution of a cylinder build was captured by the system. The technique offers strong potential for better imaging crystallisation mechanics inside the powder-cake, although this promise has yet to be realised.

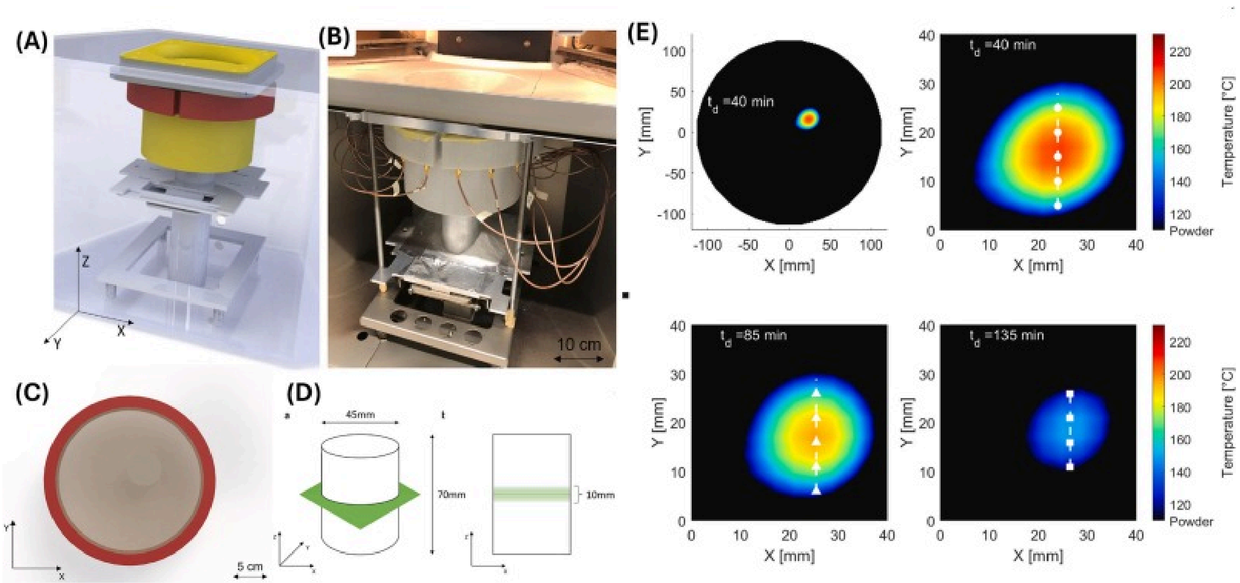

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Fig. 17. Microwave monitoring of powder cake temperature: (A) CAD model, retrofitted antenna in red, build piston in yellow, (B) actual prototype, (C) measured slice, note off-centred specimen, (D) measured slice with distribution, (E) internal powder cake temperature measurements at different time-steps and zoomed [162] (open-access article CCBY 4.0). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

5.4. Charged couple device sensors

Multispectral Charged Couple Device (CCD) imaging typically uses regular silicon CCDs to measure two or more discrete wavelengths. If the emissivity is unchanged between the two selected wavelengths, the ratio of the emissions can be used to determine the temperature of a given object. This assumption typically requires the two selected wavelengths to be very close together; however, it is not a reliable assumption for polymeric materials limiting its application in polymeric PBF. Zauner et al. [168] demonstrated temperature visualisation with an error of ±10°C from as low as 350°C with sufficient processing. For known emissivity, they demonstrated temperatures as low as 250°C were sufficiently above the CCD's noise floor to detect emissions. Key limitations of this approach is the low spectral radiance in the useful wavelengths' silicon CCDs can detect, approximately 400–1100 nm [82], [91]. Lower temperature objects have minimal emissions in this range, thus limiting the technique's applicability to polymeric powder bed applications where the melting temperature is low. Xing et al. [169] demonstrated CCD transient temperature testing with a three-wavelength ratio pyrometry approach. Decomposition of a colour image from an AOS X-PRI, 800×600, 1000 FPS high speed camera into R, G and B images was done to evaluate transient temperature fields ranging from approximately 140°C to 240°C. The signal to noise and uncertainty of this approach is not discussed, and no subsequent examples of this were apparent in literature. Accordingly, usage of CCD sensors for thermal monitoring is highly unrecommended and most likely unfeasible in polymer applications with current technology.

5.4.1. Thermal infrared imaging + laser profilometry

Combined thermal imaging and laser profilometry enables comparison between thermal distributions and the resultant spatial defects [26]. Klamert et al. observed warpage in line with the compressive and tensile stresses proposed by [62]. Sillani et al. [14] also used a Keyence LJ-X8060 laser profilometer on a DTM Sinterstation 2000 to measure powder layer quality, layer thickness, powder layer density and curling of the sintered layers. Although this study evaluated the impact of inhomogeneous thermal fields, the actual thermal field itself was not measured. Usage of laser profilometry in conjunction with thermal imaging is a highly promising area of investigation however, as it lets the user view deformation from thermal fields of surface layers in real-time. This has exciting potential for validation of simulation work, enabling simultaneous validation of both thermal and deformation simulation.

6. Recommendations

Recommendations revolve around suitable technologies for a given data acquisition target. Specifically, discussion of the temporal resolution, spatial resolution, spectral sensitivity, and cost. Fig. 18 (A) shows the key differences between microbolometer and photonic based sensors, while Fig. 18 (B) visualises the potential of microbolometer based imaging in polymeric PBF.

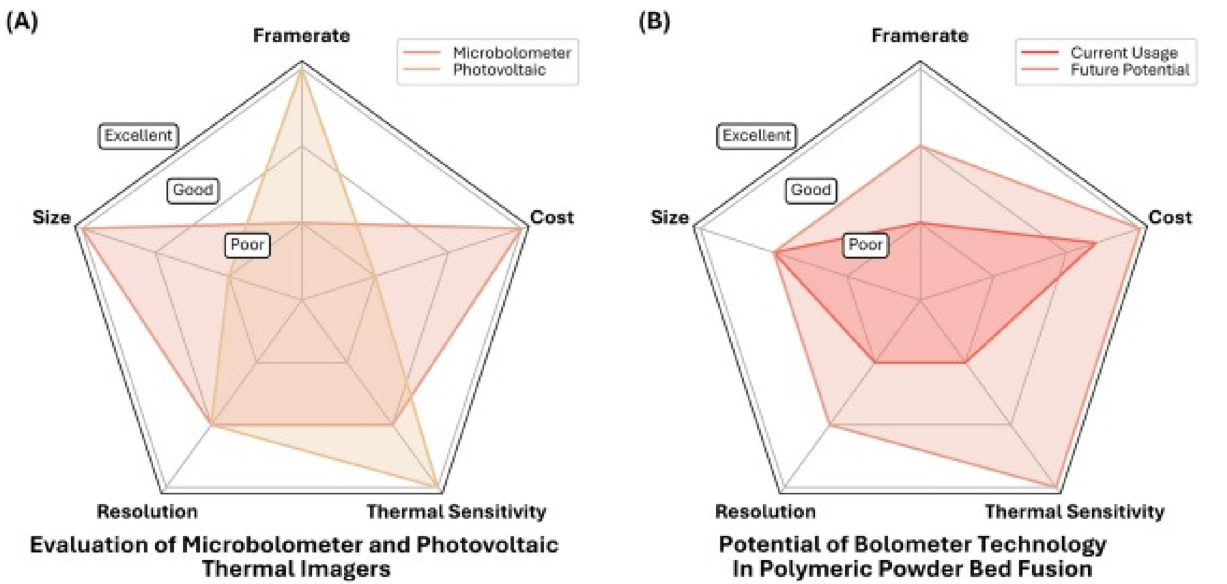

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Fig. 18. Evaluation of sensors and future potential, (A) Comparison of Microbolometer and Photovoltaic sensors, (B) Comparison of historical and potential usage of Microbolometer

sensors.

6.1. Budget thermal monitoring

Where cost is of utmost priority, K-type and T-type thermocouples are the most accessible albeit difficult to install. Distribution across the build plate, and/or insertion during the build process enables some data to be collected, with the caveat the design of experiment may be influenced by the thermocouple placement. Embedding them into the build chamber, if possible, would be one way to collect boundary conditions of the powder cake.

Another alternative to thermocouples is infrared pyrometers. Non-contact and readily integrated into most systems, they may be used with remote sensing optical heads or viewing the powder bed directly. Some models may connect via USB, though the exact output will vary. Higher frame rates and accuracy are achievable compared to focal plane array microbolometers, though this is seldom necessary. Many of the limitations faced by thermal imagers regarding emissivity apply to pyrometers, limiting their usefulness. The quantity of data to process is much lower than for thermal imagers.

Due to the strong limitations of thermocouple placement and the limited spatial resolution, low-cost microbolometers sensors such as the FLIR Lepton 3 (160×120 pixels), or the Infiray Tiny1-C (256×192 pixels) may be used. These platforms have numerous limitations such as relatively low accuracy (typical around 50 mK), aliasing issues, and low frame rates, however, offer many more data points of the build. Uncooled, they also offer a much smaller package versus more accurate and costly sensors. Applications such as full-build imaging for thermal simulation validation, sub-build imaging to evaluate specific geometric conditions, or even simply measuring bulk powder temperature are possible for below USD $1000. It should be noted sufficient reflection of laser light into the sensor may induce damage, specifically for CO_2 lasers which operate around 10 μm, within the spectral sensitivity of 7.0–14 μm seen on typical VOx microbolometers. Microbolometers are highly suited to processes like HSS and MJF due to the monitored wavelength being outside the expected wavelength of the NIR/SWIR fusing lamps. The price to performance ratio of microbolometers has steadily improved over recent years, increasing both accessibility and functionality.

One approach would be integration of thermocouples to the build plate and a microbolometer on the roof of a printer, enabling low resolution 2D imaging of the build thermal fields. In conjunction with infrared pyrometers OEM machines typically employ for laser control, characterisation of the millimetric and real scale behaviour of builds at relatively low cost is possible.

6.2. Value optimised thermal monitoring

Macroscopic thermal fields are often sufficient to determine mechanical properties and deformation of parts. Though local overheating may not be identified as clearly at lower frame rates, the longer duration of particle coalescence due to the higher melt viscosity [17], [39] diminishes the importance of high frame rate for most applications. Accordingly, higher frame rate systems typically benefit either melt pool analysis, or for feedback control of laser sintering source. Systems such as LAPS, HSS and MJF do not benefit from high-frame rate sensors as the associated energy density is substantially lower, resulting in much more gradual heating vs SLS.

Regarding cooled sensor selection, InSb sensors have higher yield than HgCdTe sensors alongside better device stability, making them a lower cost and popular option for MWIR (~3.0–5.0 μm) applications [170]. Lower cost MWIR sensors may be of minimal benefit over good quality microbolometers, thus it is recommended this category also uses microbolometer based imagers. Much of the recommendation remains the same as with budget imaging, however consideration of multiple microbolometers, potentially with optical zoom to view smaller features may be worthwhile.

Where cost allows, a singular, high-resolution MWIR sensor may be a worthwhile investment. Such a system can be set up for full-frame imaging to determine macroscopic thermal fields, then sub-framed to view identified regions of interest in lower resolution with much greater frame rate.

6.3. Peak performance

Where budget is of little to no concern, usage of multiple thermal imagers (or high-frame rate pyrometers) in conjunction with embedded thermocouples offers the most information currently feasible. This category of hardware is predominantly relevant to SLS, where the energy density (and thus heating) exhibits highly transient behaviour.

6.3.1. Scanned infrared pyrometry

Use of high-frame-rate infrared pyrometers on a separate scanning galvanometer may be beneficial for visualizing highly transient behaviour, with the viewable area adjustable according to user requirements. One limitation of this approach is the spot-size of the selected pyrometer. If the spot size exceeds the size of the region of interest, the measured temperature may be significantly lower than the actual geometry due to the sensor averaging the temperature with peripheral, lower temperature powder. Accordingly, it would be prudent to determine the degree of inaccuracy this approach may introduce by considering the projected spot size against the expected feature size.

6.3.2. Multiple thermal imagers

Use of multiple thermal imagers is another technique for comprehensive data collection. Use of a bore-sighted, high-frame-rate thermal imager coaxial to the sintering laser (sharing the galvanometer) can monitor the melt pool morphology with good temporal resolution using sub-framing to increase the frame rate. Secondary fixed imagers can be used to collect overall build plate thermal fields for evaluating millimetric and real-scale behaviour. Correlating these two data streams has been demonstrated to be highly effective at capturing the thermal field of builds.

7. Research gaps and future directions

Despite the numerous benefits of collecting thermal data during the build process, challenges and difficulties exist. In-situ monitoring is essential to allows standards to develop, the field currently lacks a systematic review. Although the underlying processes of relevance to thermal monitoring have been explored, but underlying technologies have not. In-situ thermal monitoring is critical to enabling AM to enter fields where precision, repeatability and predictability is essential such as Aerospace, Biomedical and Automotive engineering where certification and qualification is a major issue preventing wider adoption. The following gaps and future directions are identified as key impediments towards future direction.

7.1. Emissivity shifts

Temperature and phase of a powder can have large associated transmission, reflectance, and emissivity changes. Non-contact thermal monitoring depends upon accurate emissivity numbers when deriving absolute temperature. Typical approaches to address this at higher temperatures, such as multiple-wavelength approaches, are unadvisable at lower temperatures. Though numerous studies of temperature dependent emissivity exist for Polyamide powders, minimal literature exists discussing its impact upon more niche polymeric materials adding greater complexity when studying alternative or niche polymers. Thermo-optical parameters for several materials of interest are largely unexplored, for instance CB filled PP, and TPU used in HSS and MJF. Future research into this area is expected to enable more accurate thermal simulations of radiation modifying print processes.

7.2. Observer effect

Modification of existing machines to add thermal monitoring affects the underlying thermal fields of interest some extent, impacting comparison to unmodified machines. This is exacerbated for more invasive changes such as addition of thermocouples, due to the higher thermal conductivity of the probe, wire, and any potentially used spacers, thermal fields can be disrupted. Quantifying the change modification of OEM equipment has upon thermal fields is difficult but necessary when aiming for accurate data collection. Moreover, no accessible way to empirically determine internal powder-cake temperatures of production builds real-time exists. MWT while promising, remains a technology demonstration. Non-invasive powder-cake temperature monitoring does not appear to have many viable concepts to further consider. Accordingly, multi-input simulation of the internal temperatures from powder characterisation and surface data remains an interesting but minimally explored area of investigation. Novel processes such as HSS and MJF, though possessing some literature discussing internal thermal fields, omits methodology. Deeper discussion of how thermal fields behave in the sub-surface regions remains largely unexplored. One promising direction in this area is multi-input thermal field estimation with ex-situ crystallinity validation.

7.3. Data processing

High resolution and/or frame rate thermal imaging produces extremely large amounts of data. The high computational cost of processing the data limits the potential usability of the dataset. Utilisation of ML is one direction currently under investigation, with physics informed modelling of particular interest in both data processing of existing data sets and parameter extraction, as well as for predictive control. As an area of investigation, ML presents research opportunities for faster processing and greater utilisation of existing data sets.

7.4. Cost

Usage of thermal imaging in polymeric PBF predominantly focuses on high-cost cooled detectors, often on custom research machines, making recreation of results more difficult. OEM fitment of sensors is often difficult to access, necessitating modification for most users. Uncooled microbolometer sensors used are typically older models, with newer, higher-performance, and lower-cost models not currently seen in the literature e.g., FLIR Boson, Infiray P2 Pro. Opportunities exist for compact, remote datalogging retrofitted non-destructively to existing machines, with the ability to monitor both macroscopic and mesoscopic regions of arbitrary machines.

These challenges continue to exist due to the associated difficulty and cost of addressing them. Despite this, sensor and post-processing improvements continue to drive down the entry cost of monitoring, especially with newer, consumer microbolometers. Accordingly, much of the difficulty in this area has decreased, and will likely continue to do so as the consumer thermal monitoring industry grows.

8. Conclusion

In-situ thermal monitoring is beneficial for numerous applications, ranging from real-time process control, investigation of the underlying mechanics of sintering, or for process qualification and certification. In the context of polymeric PBF, the specific implementation of thermal monitoring is highly variable, not only does the application of the user matter, but so too does the subset of polymeric PBF in question. SLS, the oldest of the PBF family, is characterised by rapid, high input laser sintering of the powder with scanned passes. This introduces highly transient melt pool behaviour, detailed investigation of which has driven installation of highly expensive photonic sensors onto research machines up to now. Juxtaposed against this, newer processes such as LAPS, HSS and MJF sinter much of the layer simultaneously at a much slower rate, functionally eliminating this highly transient behaviour. These changes also have subsequent impacts on the produced part; semi-crystalline polymers' degree of crystallinity is largely driven by the thermal history of the powder and part.

The homogeneity of the thermal field, the degree of crystallinity, and the temperatures reached all drive various defects associated with AM of polymeric materials. Thermal bleed occurs due to powder fusing unintentionally to the part from excess part heat, and surface sinking occurs due to geometric conditions and shrinkage rates. Mechanical properties are largely driven by crystallinity and porosity of the manufactured part, both are highly influenced by the degree of heating and rate of cooling; due to the high viscosity of molten polymers, it can be beneficial to heat a polymer as far as possible prior to onset of degradation to reduce this viscosity as much as possible, promoting full consolidation and compaction. It should be noted excess temperature may result in varying changes to the base polymer, this is often characterised by reduced printability upon reuse, or outright decomposition of the polymer if heated sufficiently high. Reduced porosity, or greater densification improves mechanical properties through an increase in the stressed area, reducing peak stress.

Higher crystallinity is associated with polymer chains aligning with each other, resulting in a reduction in volume and yielding shrinkage. Crystallinity can also drive shrinkage, warpage, and curling, geometric distortions caused by inhomogeneous thermal fields and resultant variable rates of shrinkage by regions in a singular part. This impacts the geometric and dimensional accuracy of the part, potentially impeding correct function of a part in an assembly.

In-situ thermal monitoring encompasses numerous techniques, most relevant being thermal imagers, infrared pyrometers, and thermocouples. Ubiquitous and cheap, thermocouples can collect absolute temperature but require careful consideration of placement to prevent creation of inadvertent changes to the monitored thermal field. Pyrometers are broadly applicable, measuring a single fixed spot, low-cost models often utilising a microbolometer sensor, and more expensive models, photonic sensors which may be tuned to a specific wavelength band. Pyrometers are typically utilised by OEMs for closed-loop control of the sintering energy source.

The majority of discussion in this paper focused on thermal imagers, despite the higher cost compared to alternative systems, they can collect spatial data and can be retrofitted to machines less invasively than alternatives like thermocouples. Thermal imagers, much like pyrometers, rely upon the blackbody radiation emitted by heated bodies, however the mechanism through which they detect this varies. Microbolometer type sensors measure the temperature dependent resistivity of the bolometer membrane (typically VOx or α-Si), which is heated up by incident radiation. These sensors have limited frame rate due to the thermal mass of the bolometer pixel, as well as poorer thermal sensitivity compared to photonic-type detectors. Photonic type detectors are traditional high-cost, high-performance scientific detectors, characterised by higher sensor gain. This comes at the cost of the sensor relying cryogenic cooling (often near 80K) to operate. Photonic type sensors use photovoltaic sensors (though photoconductive can also work) to collect photons. These photons are collected for the duration of the user set integration time, then pixel values are read out through the ROIC to determine the temperature of the imaged object. Due to this the achievable frame rate exceeds microbolometer type sensors by two orders of magnitude.

Mounting of the two sensors, as well as maintaining suitable conditions for clear imaging adds some degree of complexity, however. Large angles of incidence (typically more than 30°, or greater than the manufacturer's specification) can reduce the measured temperature. Sensors should be thermally isolated from the build chamber which operates at temperatures in excess of 100°C, however condensation of vapour from sintering processes can occlude the window necessitating further design such as nitrogen gas shielding, or intentional heating of the window to prevent condensation. Furthermore, all non-contact sensors are highly dependent upon knowing the target emissivity, which can fluctuate unpredictably as temperature rises, and especially so during phase transitions from solid, to liquid, to solid.

Novel approaches discussed were also particularly interesting, use of laser profilometry to quantify curling numerically against measured thermal fields has strong implications for validation of thermo-mechanical simulation studying warpage of printed parts. Microwave Tomography as a concept was also a promising demonstration for how internal powder-cake temperature could be collected, although this was greatly limited in scope as a technological demonstration.

Evaluation of current academic usage of in-situ thermal monitoring in polymeric PBF found photonic sensors, specifically InSb sensors highly prevalent due to the high thermal sensitivity, frame rate, and good resolution. Often machines with InSb sensors were custom-built, this can be explained due to the bulkier packaging of the cooled sensor making it more difficult to retrofit to existing machines without prior provision for monitoring addons. Retrofitted setups, or minimal test setups (e.g. single layer demonstration) often used microbolometer type sensors, this can be attributed to the smaller package of the sensor which lacks cooling. Overall recommendations for the interested reader were made, for most users newer microbolometer sensors with appropriate thermal management suffices. The higher frame rate enabled by InSb sensors are typically only required for melt pool analysis in SLS, a property newer, less understood processes like LAPS, HSS and MJF do not require.

Despite the immense design freedom enabled by AM, it is evident numerous challenges exist providing a robust, reliable, and predictable outcome to generalised geometries and processes. Encouragingly, numerous areas of inquiry once difficult for cost reasons are increasingly open for exploration due to development in sensor technology and wider consumer markets. Investigation into the identified gaps more importantly enables us to move towards reduced wastage and greater efficiency in manufacturing. These goals can only be achieved through sufficient understanding and characterisation of processes to pursue process qualification by governing bodies, something in-situ thermal monitoring can greatly help with at every step.

CRediT authorship contribution statement

Han Yu Ooi: Writing – original draft, Visualization, Data curation. **Jordan Noronha:** Writing – review & editing. **Tristan Pon:** Writing – review & editing. **Martin Leary:** Writing – review & editing. **David Downing:** Writing – review & editing. **Eric MacDonald:** Writing – review & editing. **Stuart Bateman:** Writing – review & editing, Funding acquisition. **Raj Das:** Writing – review & editing. **Mahyar Khorasani:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table A. Papers utilising thermal cameras, pyrometers, or thermocouples.

References	Polymer	IR Imager	IR Imager	Thermocouple	Pyrometer	Other
[171]	Fine Nylon LNF5000	×	×	✓	✓	×
[39]	EOS PA2200 PA12	TIM 400	×	×	✓	×
[162]	Duraform PA12	×	×	×	×	MWT
[15]	HP PA12	×	×	✓	×	×
[26]	EOS PA2200 PA12	FLIR T420	×	×	✓	Laser Profilometer
[98]	EOS PA2200 PA12	InfraTec ImageIR 8300	×	×	×	×

References	Polymer	IR Imager	IR Imager	Thermocouple	Pyrometer	Other
[172]	BASF PP Nat 01	IRCAM VELOX 1310k	×	×	×	×
[173]	EOS PA2202 PA12 CB	IRCAM VELOX 1310k	×	×	×	×
[174]	EOS PA2200 PA12	IRCAM VELOX 1310k	×	×	×	×
[175]	EOS PA2200 PA12	IRCAM VELOX 1310k	×	×	✓	×
[176]	BASF Ultrasint PP 1400	IRCAM VELOX 1310k	×	×	×	×
[35]	EOS PA2201 PA12	×	×	×	✓	×
[130]	Axalta Vulcan Black, Axalta Oil Black, TIGER Drylac TD61 series Anodized Effect	FLIR X6901sc	×	×	×	×
[177]	LyondellBasell isotactic homo Polypropylene, Evonik Polyamid12 PA12	IRCAM VELOX 1310k	×	×	✓	×

References	Polymer	IR Imager	IR Imager	Thermocouple	Pyrometer	Other
[57]	ALM PA 650 PA12	FLIR SC7600	×	×	×	×
[12]	PA12	FLIR A325	×	×	×	×
[178]	ALM PA 650 PA12	FLIR SC7000	×	×	×	×
[164]	✓	×	×	×	✓	×
[37]	ALM PA 650 PA12	FLIR 6701sc	FLIR SC8243	✓	×	×
[38]	Duraform PA PA12	FLIR A325	×	✓	×	×
[179]	EOS PA2200 PA12	IRCAM VELOX 1310k	×	×	×	×
[105]	PA12 CB	Seek Compact	×	✓	×	×
[29]	✓	×	×	×	✓	×
[140]	EOS PA2200 PA12	IRCAM VELOX 1310k	×	×	×	×
[41]	ALM PA 650 PA12	FLIR A6701	FLIR SC8240	×	×	×
[180]	PA 12	FLIR A6701	FLIR SC8240	×	×	×
[42]	ALM PA 615 GS PA12 GB	FLIR Lepton 3.5	×	×	×	×

References	Polymer	IR Imager	IR Imager	Thermocouple	Pyrometer	Other
[181]	EOS PA2200 PA12	Unspecified	×	×	×	×
[182]	EOS PA2200 PA12	×	×	✓	×	×
[183]	Copolymerised Polylactide	FLIR A600 series	×	×	×	×
[45]	Shanghai TOMIS Material Technology Co., Ltd. PES-HmA	FLIR Ti400	×	×	✓	×
[142]	EOS PA2200 PA12	TIM 400	×	×	×	×
[67]	EOS PA2200 PA12	Velox 1310k SM	×	×	×	×
[75]	PA12	FLIR A325	×	×	×	×
[58]	ALM PA650 PA12	FLIR SC7000	×	×	×	×
[159]	PA12	FLIR A325	×	×	×	×
[148]	Precimid 1171 PA12	Testo 890–2	×	×	×	×
[184]	PA12	A6701 FLIR	×	×	×	×
[185]	ALM PA 650 PA12	FLIR A6701	FLIR SC8240	×	×	×

References	Polymer	IR Imager	IR Imager	Thermocouple	Pyrometer	Other
[151]	DuraForm™ PA12	FLIR E40	×	✓	×	×
[153]	Sinterit PA12 w/ Nanesa S.r.l GnP powder	Fotric 348A	×	×	×	×
[155]	Dalian Sanben Casting Co Spherical Fused Ceramsite with phenolic resin	DT-980, Huashengchang Technology	×	✓	×	×
[40]	ALM PA 650 PA12	FLIR A6701	FLIR SC8240	✓	×	×
[163]	SolidConcepts PEEK	FLIR A325	×	K-Type	×	×
[157]	sPS XAREC S90Z (Modified)	Testo 875	×	×	×	×
[186]	EOS PA2200 PA12	IRCAM VELOX 1310k	×	×	✓	×
[160]	EOS PA2200 PA12	×	×	✓	×	×
[69]	Coated Sand	FLIR A310	×	×	×	×
[137]	PA2200	InfraTec Jade III	×	×	×	×

References	Polymer	IR Imager	IR Imager	Thermocouple	Pyrometer	Other
[187]	DTM Laserite PC Compound LPC-3000 with Riedel-de Han graphite fine powder 15,553	×	×	×	Mikron M67S	×
[32]	NA	Unspecified	×	✓	×	×
[133]	EOS PA12 Black PA2202 White PA2200 and	Infratec ImageIR 5300	×	4× Wireless	×	×
[167]	Unknown mixed PA powder	×	×	×	microIRt/c Exergen	×
[128]	PA12	Flir T420	×	×	×	×
[188]	EOS PA12, PA2202	FLIR A325sc	×	×	×	×
[189]	EOS PA2200 PA12	IRCAM VELOX 1310k	×	✓	✓	×
[44]	ALM PA250 PA12	Micro Epsilon TIM640	×	✓	×	×
[136]	EOS PA2200 PA12	InfraTec ImageIR 8300	×	×	×	×

References	Polymer	IR Imager	IR Imager	Thermocouple	Pyrometer	Other
[190]	EOS PA2200 PA12	IRCAM VELOX 1310k	×	×	×	×
[73]	EOS PA2200 PA12 w/ 1% CB, Dupont PD0580 PP w/ 1% Evonik Aerosil R106 & 1% CB	Unspecified	×	✓	×	×
[17]	PA 12, PEKK	FLIR SC4000	×	×	✓	×
[30]	EOS PA2200	×	×	K-Type	✓	×
[36]	Corbion PURASORB PDLG 7507 PLGA	FLIR A600 series	×	×	×	×
[145]	Sintratec PA12 CB, Ultrasint PA6 CB	Optris Xi400	Seek Thermal Compact	✓	×	×
[134]	Sintratec PA12 CB	Infratec ImageIR 5385	×	×	✓	×
[191]	PA12	Unspecified	×	×	✓	×
[59]	✓	Unspecified	×	×	×	×
[192]	TM (Acrylic/Styrene CoP)/SiO2	×	×	×	Mikron M67S	×

References	Polymer	IR Imager	IR Imager	Thermocouple	Pyrometer	Other
[193]	EOS PA2202	SC4000	×	×	×	×

Data availability

Data will be made available on request.

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Han Yu Ooi is a PhD Candidate at the RMIT University's Centre for Additive Manufacturing, focusing on process modelling of Multi Jet Fusion of Polypropylene parts. He graduated from RMIT University with a Bachelor of Advanced Manufacturing and Mechatronics alongside a Bachelor of International Business. Other areas of professional work have included research and development of medical actuator prototyping, as well as ongoing engagements as a Casual Researcher in mechanical design, manufacture and assembly of Power Electronic converters.

Jordan Noronha (PhD) is an Early Career Postdoctoral Research Fellow at RMIT University's Centre for Additive Manufacturing. In his PhD, he pioneered the laser-based powder bed fusion (PBF-LB) fabrication of microtubes, developing complex lattice architectures known as hollow-strut lattices. Dr. Noronha now supervises PhD students and lectures in design for additive manufacturing (DFAM) principles, metal AM processes, and artificial intelligence, while continuing his research in developing stronger, lighter metamaterials.

Tristan Pon is a Graduate Corrosion Engineer and a PhD candidate at RMIT University, focusing on the applications of machine learning in Additive Manufacturing, with a particular emphasis on Multi Jet Fusion and Polypropylene feedstocks. Tristan graduated from RMIT with a Bachelor of Mechanical Engineering with First Class Honours.

Martin Leary began his scholarly career as a postgraduate student within the Engineering Design Group at the Mechanical Engineering Department of the University of Melbourne. As a member of the RMIT School of Aerospace Mechanical and Manufacturing Engineering, he proudly continues to engage with industry and develop novel research outcomes within four research fields: Engineering design and engineering education, Systems approaches and optimisation, Sustainable automotive system design, and Renewable energy.

David Downing is a Research Fellow at RMIT University's Centre for Additive Manufacturing with an interest in Design for Additive Manufacturing, process monitoring and numerical modelling. David studied robot kinematics at the University of Melbourne and previously developed novel bottle design with packaging company Amcor.

Eric MacDonald received his doctoral (2002) degree in Electrical and Computer Engineering from the University of Texas at Austin. He worked in industry for 12 years at IBM and Motorola and subsequently co-founded a start-up specializing in CAD software and the startup was acquired by

a firm in Silicon Valley. Dr. MacDonald spent 2003 to 2016 at the University of Texas at El Paso as the associate director of the W. M. Keck Center for 3D Innovation and held faculty fellowships at NASA's Jet Propulsion Laboratory, US Navy Research and was awarded a US State Department Fulbright Fellowship in South America.

Stuart Bateman, Head of Department and Associate Dean Mechanical, Manufacturing and Mechatronics Engineering, Professor Bateman is an active researcher in high performance and functional polymeric materials and coatings. He focuses on solving applied research problems which underpin competitive advantage to partners and longer-term impact. Prior to joining RMIT in 2015, he spent 17 years with CSIRO as a researcher and research leader. Key roles included being the Theme Leader of CSIRO Sustainable Polymeric Materials (50 staff, $18 M research portfolio) from 2005 to 2011, and the Deputy Director of the Future Manufacturing Research Flagship from 2011 to 2014 (a 300 staff, $100 M research portfolio).

Raj Das is the Full Professor of 'Structures and Materials' and leads the 'Simulation of Advanced Materials and Structures (SAMS)' research group in the 'Sir Lawrence Wackett Research Centre' of RMIT University. He is the 'Program Director' in the School of Engineering's Aerospace Engineering and Aviation discipline. He is also an honorary academic in the University of Auckland, New Zealand, and the University of Quebec, Canada. He has nearly 20 years of experience in the design, analysis and optimisation of engineering materials and structures, with a focus on computational mechanics, structural optimisation, composite structures, failure analysis, and damage tolerance design.

Mahyar Khorasani works at RMIT as Research Officer at the Centre for Additive Manufacturing. He is the co-author of a leading publication in the field. Mahyar is Associate Editor in Chief of Rapid Prototyping Journal which was the first established journal in this field. Current research aims to identify physical principles to improve a temperature-dependent optical-model in 3D-printing of Multi Jet Fusion Polypropylene which expects to generate new knowledge using an innovative approach on process temperature estimation. This should provide significant benefits, such as the possibility to predict the quality of the printed components before starting the print jobs, and the potential to grow manufacturing in Australia.

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